

Full Length Article

Unsupervised motion artifact purification guided by joint prior from pixel and k-space domains

Jiahua Xu^{a,1}, Dawei Zhou^{b,1}, Lei Hu^{c,1}, Jianfeng Guo^d, Feng Yang^e, Zaiyi Liu^c, Nannan Wang^{a,*}, Xinbo Gao^a

^a State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an, 710071, Shaanxi, China

^b Faculty of Data Science, City University of Macau, Macao Special Administrative Region, 999078, China

^c Department of Radiology, Guangdong Provincial People's Hospital, Guangzhou, 510080, Guangdong, China

^d Department of Radiology, Huangshi Central Hospital, Affiliated Hospital of Hubei Polytechnic University, Huangshi, 435000, Hubei, China

^e Department of Radiology, Xiangyang No. 1 People's Hospital, Hubei University of Medicine, Xiangyang, 441000, Hubei, China

ARTICLE INFO

Keywords:

Unsupervised learning
Image generation
Medical imaging enhancement
Motion artifact removal
Diffusion model
Joint prior guidance

ABSTRACT

Magnetic resonance imaging (MRI) plays an important role in clinical diagnosis, nevertheless, it suffers from motion artifacts in many real-world scenarios. Removing motion artifacts can alleviate this interference and thus facilitate precise diagnosis. However, recent works mainly rely on paired data and the artifactual perturbations in k-space (frequency domain) have not yet received sufficient attention. In this work, we propose an unsupervised purification method which incorporates k-space information of noisy MRI images as a guidance to acquire clean MRI images via a pre-trained diffusion model. Specifically, motion artifacts are mainly carried by high-frequency components in k-space, so the low-frequency components can be utilized as guidance to recover proper tissue texture. Meanwhile, considering that high-frequency and pixel information are still valuable for refining shape and texture details, we design alternate complementary masks to leverage useful information while destroying their artifactual structure. Additionally, we propose a prior meanbook as a stable denoising reference, which effectively improves the performance of artifact removal. Experiments on three datasets from different body parts are conducted to evaluate the effectiveness of the proposed method, which show that our method exhibits superior performance in multiple quantitative metrics and the qualitative clinical assessments.

1. Introduction

As a non-invasive medical imaging technique, magnetic resonance imaging (MRI) has been widely applied in healthcare and plays an important role in clinical diagnosis. However, its scanning usually takes extended acquisition time (Zaitsev et al., 2015). During the scanning process, some patients, such as elderly or those with conditions such as claustrophobia and epilepsy (Chen, 1990), may experience movements due to discomfort or uncontrollability. These movements may disturb the k-space (also known as frequency domain) signal distribution of MRI, resulting in motion artifacts and further seriously interfering with clinical diagnosis and treatment. Therefore, how to effectively purify motion artifacts to mitigate their affects is a significant task, which is what this paper is working on.

Previous several researches have made explorations to solve this problem (Hu et al., 2024a,b; Schulz et al., 2012; Zaitsev et al., 2006).

They are mainly categorized into prospective motion artifact prevention and retrospective motion artifact purification. The former is dedicated to proactively avoiding the generation of motion artifacts based on tracking the motion (Ehman et al., 1984; Qin et al., 2009; Schulz et al., 2012; Zaitsev et al., 2006), such as by adjusting imaging sequences and parameters (Block et al., 2007; Cruz et al., 2016, 2017; Vasanaawala et al., 2010b; White et al., 2009). However, such strategies usually require additional equipment or longer scanning times, leading to higher costs. The latter aim to iteratively correct motion artifacts using compression perception (Hansen et al., 2012; Jin et al., 2017; Küstner et al., 2017; Vasanaawala et al., 2010a; Yang et al., 2013). However, such strategies typically need raw k-space data that are difficult to access, and this data barrier poses a challenge to efficiently construct models. Besides, traditional approaches rely heavily on paired data for fully supervised training (Isola et al., 2017; Tamada et al., 2020), but it is difficult and costly to obtain precisely paired data due to the movement of body parts and

* Corresponding author.

E-mail addresses: jhxu.xidian@gmail.com (J. Xu), dwzhou@cityu.edu.mo (D. Zhou), hulei@gdph.org.cn (L. Hu), jianfengguo@hbmh.edu.cn (J. Guo), haitang76@163.com (F. Yang), liuzaiyi@gdph.org.cn (Z. Liu), nnwang@xidian.edu.cn (N. Wang), xbgao@mail.xidian.edu.cn (X. Gao).

¹ These authors contributed equally to this work.

<https://doi.org/10.1016/j.neunet.2026.109573>

Received 21 February 2026; Received in revised form 15 June 2026; Accepted 28 August 2026

Available online 5 September 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

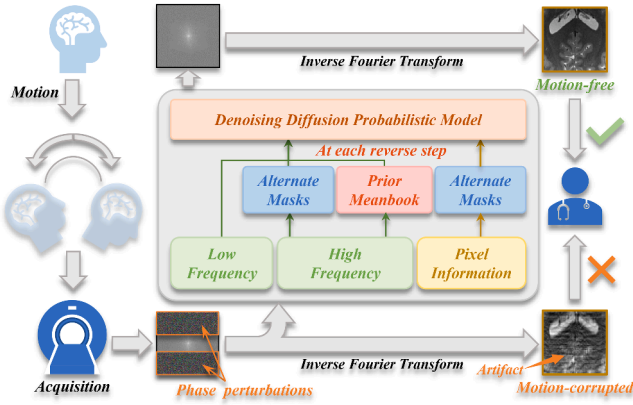


Fig. 1. Schematic of our research problem and solution. During the MRI imaging process, motion artifacts are produced due to the patient's motion. Removing artifacts via a diffusion model with prior meanbook and alternate masks can help radiologists make better diagnosis.

the natural motion of organs during rescanning. Meanwhile, various machine learning models have shown strong ability in modeling complex nonlinear patterns, such as neural networks (Jin & Xu, 2025a,b), Gaussian process regression (Jin & Xu, 2024, 2025c), graphical techniques (Jin & Xu, 2025d, 2026), ensemble methods (Xu, 2020; Xu & Zhang, 2021a), and network-based techniques (Xu & Zhang, 2021b, 2023). However, MRI motion artifacts are usually related to structured k -space perturbations, which are difficult to directly model and correct using general machine learning methods. Therefore, generative models with progressive refinement ability are more suitable for this task, especially diffusion models.

Recently, deep learning methods (Zhang et al., 2025a,b,c) and denoising diffusion probabilistic model (DDPM) has shown excellent performance in image generation (Dhariwal & Nichol, 2021; Ho et al., 2020; Ho & Salimans, 2022; Rombach et al., 2022; Saharia et al., 2022a; Song et al., 2020). It has been shown that integrating images (Dhariwal & Nichol, 2021; Rombach et al., 2022; Saharia et al., 2022a) or text (Gao et al., 2024; Ho & Salimans, 2022; Nichol et al., 2021; Ramesh et al., 2022; Saharia et al., 2022b; Si et al., 2023; Tumanyan et al., 2023; Zhang et al., 2023) as guidance in the reverse process of DDPM can promote it to generate desired images according to specific requirements. However, although diffusion-based approaches perform well for reconstruction or purification of natural images, they still exhibit limited ability in purifying motion artifacts for MRI images. The reason for this issue is that motion artifacts mainly originate from specific components in k -space, whereas these approaches only focus on pixel-domain information during their purification process.

To address above issues, we aim to purify motion artifacts in MRI images with an unsupervised mechanisms by jointly exploiting the prior knowledge from both pixel and k -space domain. As shown in Fig. 1, we first exploit unpaired clean MRI images to construct a pre-trained diffusion model, which *learns the data distribution and the generation mapping of MRI images without motion artifacts*. Secondly, we extract the k -space low-frequency component of the input noisy image, and take it as conditional guidance to facilitate the diffusion model to generate clean and proper tissue texture. Simultaneously, since the high-frequency component and pixel information are still valuable for refining shape and texture details, we design an alternate complementary mask mechanism for them, aiming to *leveraging their useful information while destroying the structure of motion artifacts in them*. The masks are flipped at each reverse step to alternatively cover all image areas. Following this, we introduce a parameter that changes with diffusion time to further disrupt the distribution of artifacts. Additionally, we propose a prior meanbook that clusters clean images to build a statistically stable reference prior, thereby *providing reliable guidance in the reverse process of the diffusion model and*

enabling accurate structural recovery. To demonstrate the effectiveness of our proposed method, we performed quantitative metrics evaluations and qualitative clinical assessments on three datasets covering different body parts, respectively.

The main contributions are summarized as follows:

- To mitigate the interference of motion artifacts of MRI images in the absence of paired data, we propose an unsupervised Motion Artifact Purification method called MAPPK, which constructs a diffusion model guided by prior knowledge from the pixel and k -space domains.
- Alternate complementary masks are designed for high-frequency and pixel components to suppress artifacts while preserving useful information for clean signal recovery. Two parameters dynamically adjust the ratio of guidance from the noisy and generated images, and trade off the recovered signals from the pixel and frequency domains, respectively, to simultaneously promote artifact removal and image quality.
- We introduce a prior meanbook that constructs robust reference priors from clustered clean MRI images, guiding the diffusion model during the reverse process to achieve accurate structural restoration.
- In addition to quantitative evaluation on datasets with simulated artifacts, we perform experiments on collected MRI data with different degrees of real motion artifacts, and commission multiple radiologists to perform clinical assessments.

This paper extends our previous work (Xu et al., 2025) in several aspects. 1) **Methodology:** we introduce a prior meanbook that constructs a statistically stable reference prior from clustered clean MRI images to guide the diffusion model for accurate structure recovery and incorporate residual connections to improve the inverse diffusion process, thus further suppressing the influence of artifacts. Moreover, we start the reverse process from an intermediate timestep to accelerate inference. 2) **Experiment:** we extend the evaluation to MRI data with severe motion artifacts to demonstrate the effectiveness and robustness of our method, and perform ablation studies to verify the contributions of the prior meanbook, mask, and dynamic guidance parameters. 3) **Task scenarios:** we propose to use different frequency thresholds to process complex clinical data with different degrees of real motion artifacts. Due to the flexibility of our method, we can perform this process without additional training.

2. Related work

2.1. MRI motion artifacts

Previous work (Budde et al., 2014) reported that patient motion during MRI acquisition can introduce errors in k -space along specific phase-encoding lines, leading to image blurring or ghosting artifacts. Therefore, refer to Oh et al. (2023), we adopt a lenient assumption that the motion artifacts in MRI images predominantly manifest in the high-frequency region along the phase encoding lines. It is due to the fact that k -space data acquisition usually starts from the centre region, during which the patient usually does not move spontaneously. It is only after a period of time does the patient move, resulting in perturbation of phase information in the high frequency region. Consequently, our focus lies on removing artifacts in the high-frequency region, and set it to be greater than $\pi/10$.

2.2. Motion artifact removal

Motion artifacts can significantly disrupt clinical diagnosis and treatment. With the rapid development of deep learning (Ahn et al., 2023; Dalva et al., 2022; Feng et al., 2023; Hu et al., 2024a,b; Jiang et al., 2023; Karras et al., 2019; Li et al., 2024, 2025; Liu & Fieguth, 2012; Liu et al., 2016, 2025, 2021; Wang et al., 2018; Zhang et al., 2021; Zhou

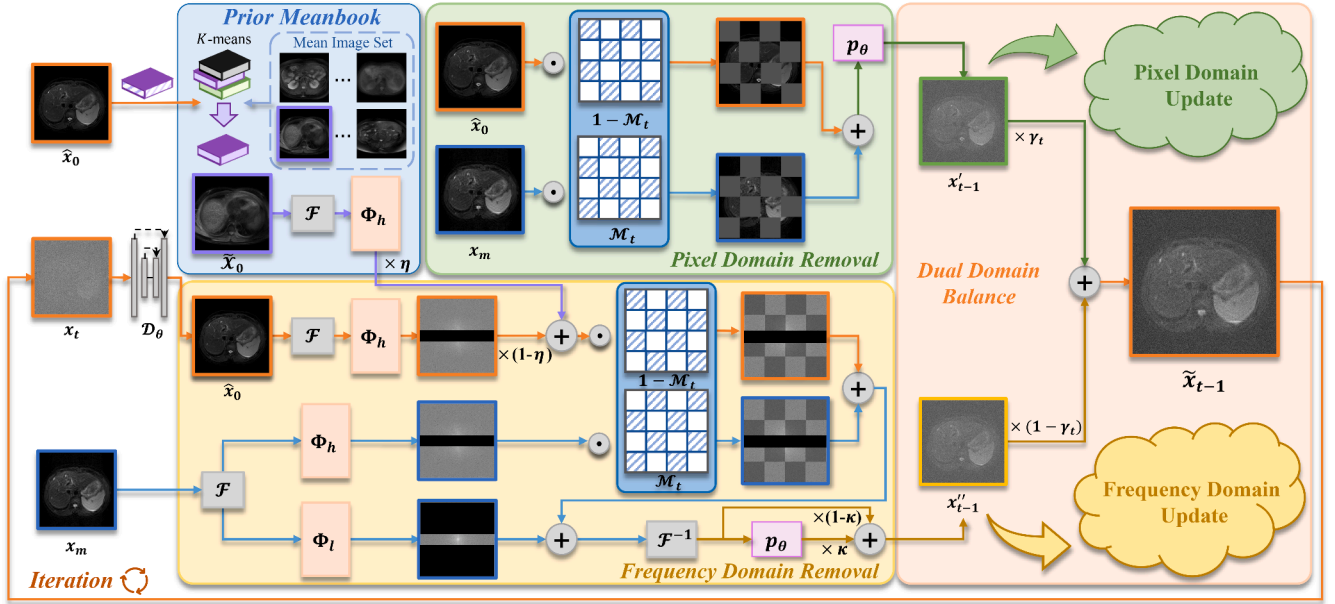


Fig. 2. The framework of MAPPK. The blue area is the prior meanbook which leverages the reference prior to guide the diffusion model to generate accurate structural textures. The yellow area is the artifact removal process in frequency domain. This process leverages the low-frequency information and part of the high-frequency information of the motion-corrupted image and reference prior to generate another part of clean high-frequency information. The green area is the pixel domain artifact removal process, which leverages the partial pixel information of the motion-corrupted image as guidance to smooth the image. After combining the both domains results, the clean image is finally generated iteratively.

et al., 2025a,b), convolutional neural networks have gradually been applied to motion artifact removal in MRI images.

Researchers used GAN-based approaches, such as Pix2pix (Isola et al., 2017), CycleGAN (Zhu et al., 2017), CycleMedGAN-V2.0 (Armanious et al., 2020) and stochastic degradation block (Chung et al., 2021) to remove artifacts. However, in reality, the distribution of motion artifacts is variable due to changes in motion forms, so these approaches are difficult to cope with complex clinical scenario. Thus, researchers have further explored diffusion models for MRI image restoration, such as DR2 (Wang et al., 2023), GDP (Fei et al., 2023), and BIRD (Chihouai et al., 2024). In addition, recent diffusion models have focused on MRI image restoration tasks, such as CCDM (Qiu et al., 2026), ABSR (Qiu et al., 2024b), and DBSR (Qiu et al., 2024a), further demonstrating the potential of diffusion models in MRI image processing. Compared with general machine learning methods (Jin & Xu, 2024, 2025a,b,c,d, 2026; Xu, 2020; Xu & Zhang, 2021a,b, 2023), diffusion models provide an iterative denoising process and can naturally incorporate conditional guidance. This makes them more suitable for progressively suppressing motion-related k -space perturbations in unsupervised MRI artifact removal.

The above approaches are difficult to apply in the clinical diagnosis due to the neglect of k -space phase perturbations. Therefore, researchers proposed UDDN (Wu et al., 2023) to remove motion artifacts in frequency domain. Meanwhile, Oh et al. (2023) used the score-based diffusion model to perform the recovery using measured k -space values. Noting that most previous works are based on GANs and remove artifacts only in pixel domain, we propose a novel method based on DDPM in pixel-frequency domain.

3. Methodology

3.1. Preliminary

3.1.1. Notice

We use lower-case bold letters $\mathbf{x} \in \mathbb{R}^{1 \times H \times W}$ to denote the image. Moreover, we use t to represent the timestep that the diffusion model is going through and T to represent the total number of timesteps, with $t \in$

$\{1, \dots, T\}$. We use $\mathcal{N}(\cdot; \mu, \sigma^2 \mathbf{I})$ to denote a variable that obeys a Gaussian distribution with mean μ and variance matrix $\sigma^2 \mathbf{I}$. We denote a diffusion model by $\mathcal{D}_\theta$, where θ is its model parameters. We use $\mathcal{F}$ to represent the Fourier transform, $\mathcal{F}^{-1}$ to represent the inverse Fourier transform, and Φ_l and Φ_h to represent a pair of complementary ideal low-pass filter and high-pass filter. We use $|\cdot|$ to represent the modulo value operation and $\odot$ to represent the Hadamard product.

3.1.2. Diffusion model

The diffusion model is a kind of generative model (Dhariwal & Nichol, 2021; Ho et al., 2020; Rombach et al., 2022; Saharia et al., 2022a; Song et al., 2020). Specifically, given an image $\mathbf{x}_0$, the forward process gradually adds noise to obtain $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$. This is represented as follows:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (1)$$

where $\beta_t \in \{\beta_1, \dots, \beta_T\}$ represents a variable that grows linearly with the timesteps. $\mathbf{I}$ is the identity matrix.

At the same time, we can also get the reverse process,

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I}), \quad (2)$$

where μ_θ denotes a trainable neural network, σ_t^2 is always set as constants, and the iteration starts from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. In this paper, we use p_θ to represent Eq. (2), which can be expressed as follows:

$$\mathbf{x}_{t-1} = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_0, \quad (3)$$

where $\hat{\mathbf{x}}_0$ is the image predicted by μ_θ .

To train the neural network, we can obtain the optimization objective as follows:

$$\mathcal{L}_{diff} = \mathbb{E}_{\mathbf{x}_0, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\|\epsilon - \epsilon_\theta(\sqrt{\alpha_t} \cdot \mathbf{x}_0 + \sqrt{1 - \alpha_t} \cdot \epsilon, t)\|^2 \right], \quad (4)$$

where ϵ_θ denotes the noise prediction network based on U-Net (Ronneberger et al., 2015) with parameter θ , $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

3.2. Motion artifact removal

The framework of our method is shown in Fig. 2. Firstly, we introduce how to build and use the prior meanbook. Secondly, we introduce how to create alternate complementary masks. Then, we introduce how to use alternate complementary masks in frequency domain and pixel domain, respectively. Finally, we design a hyperparameter to balance the weights of the two domains.

3.2.1. Prior meanbook

In order to provide a stable and representative denoising prior for the reverse process, we propose prior meanbook, which serves as a reliable reference image library in the artifact removal process and can effectively guide the diffusion model to generate clean images. The construction and selection of the prior meanbook are fully automatic and objective, without any manual intervention.

Specifically, we use K -means algorithm to divide the clean images into k categories,

$$\{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_k\} = Kmeans(\mathcal{X}_c), \quad (5)$$

where $\mathcal{X}_c$ is the set of clean images, $\mathcal{X} \in \mathbb{R}^{1 \times H \times W}$ and $\{\mathcal{X}_1, \mathcal{X}_2, \dots, \mathcal{X}_k\}$ represents k clean image subsets obtained after K -means classification. In this paper we set k to 50. In order to obtain a statistically significant prior library, we calculate the pixel-wise mean value of each subset.

$$\bar{\mathcal{X}}_s = \frac{1}{N} \sum_{n=1}^N \mathcal{X}_s^n, \quad s \in \{1, 2, \dots, k\}, \quad (6)$$

where $\bar{\mathcal{X}}_s$ represents the pixel-wise mean value of the s th category $\mathcal{X}_s$, and N is the number of samples in $\mathcal{X}_s$. Given the structural differences between slices of different patients, the mean images can introduce local information confusion. Therefore, we introduce the Laplacian filter to smooth out the mean images.

$$\tilde{\mathcal{X}}_s = LaF(\bar{\mathcal{X}}_s, \lambda), \quad (7)$$

where $\tilde{\mathcal{X}}_s \in \mathbb{R}^{1 \times H \times W}$ denotes the smoothed mean images, LaF represents the Laplacian filter, and λ is the smoothing strength, which is set to 0.1 in this paper. At each reverse iteration, after predicting $\hat{x}_0$, we compute its distance to all mean images in the prior meanbook and automatically select the closest mean image as the prior reference for the current step.

3.2.2. Frequency domain removal

We first focus on removing motion artifacts in frequency domain. The algorithm of this module is shown in Algorithm 1.

Firstly, we extract the low-frequency information from x_m to ensure anatomical structures,

$$f_l = \Phi_l(F(x_m)), \quad (8)$$

where x_m denotes the motion-corrupted image and $f_l \in \mathbb{R}^{1 \times H \times W}$ represents the low frequency information extracted from x_m .

After that, we extract the high-frequency information of x_m to guide the reverse process,

$$f_h^m = \Phi_h(F(x_m)), \quad (9)$$

where $f_h^m \in \mathbb{R}^{1 \times H \times W}$ represents the high-frequency information extracted from x_m .

Although diffusion models possess strong generative capabilities, high-frequency guidance information containing artifacts can cause each predicted $\hat{x}_0$ to deviate from the clean distribution. Therefore, we introduce prior meanbook to correct this bias,

$$f_h^r = \eta \cdot \Phi_h(F(\tilde{\mathcal{X}}_0)) + (1 - \eta) \cdot \Phi_h(F(\hat{x}_0)), \quad (10)$$

where $f_h^r \in \mathbb{R}^{1 \times H \times W}$ represents the high-frequency information reorganized from the prediction $\hat{x}_0$ of diffusion model and $\tilde{\mathcal{X}}_0, \tilde{\mathcal{X}}$ represents the mean image predicted by K -means that has the closest Euclidean distance to $\hat{x}_0$, and η is a trade-off hyperparameter.

Algorithm 1 Frequency domain removal.

Input: Current image x_t , motion-corrupted image x_m , the predicted image $\hat{x}_0$, high-pass and low-pass filters Φ_h, Φ_l , intrinsic variable of diffusion model $\bar{\alpha}_t$, the alternate complementary masks $\mathcal{M}_t$, prior meanbook $\{\tilde{\mathcal{X}}_s\}_{s=1}^k$, trade-off hyperparameter η .

- 1: Obtain the mean image $\tilde{\mathcal{X}}_0$ corresponding to $\hat{x}_0$ by K -means algorithm from prior meanbook $\{\tilde{\mathcal{X}}_s\}_{s=1}^k$;
- 2: Extract the low-frequency information f_l and high-frequency information f_h^m of the motion-corrupted image x_m using Eqs. (8) and (9), respectively;
- 3: Obtain the reorganized high-frequency information f_h^r from $\hat{x}_0$ using Eq. (10);
- 4: Calculate the final high-frequency information using Eq. (11);
- 5: Combine the reorganized high-frequency information with the original low-frequency information to obtain the reorganized image x'_0 using Eq. (12);
- 6: Obtain the next iterative image x'_{t-1} of frequency domain using Eq. (13);

Output: Updated image in frequency domain x'_{t-1} .

Then, we apply the alternate complementary masks in the high-frequency region,

$$f_h = f_h^m \odot \mathcal{M}_t + f_h^r \odot (1 - \mathcal{M}_t), \quad (11)$$

where $f_h \in \mathbb{R}^{1 \times H \times W}$ is the final reorganised high-frequency information, $\mathcal{M}_t \in \mathbb{R}^{1 \times H \times W}$ is the alternate complementary masks, which will be introduced later.

By combining the reorganised high-frequency information with the original low-frequency information, we can obtain the reorganised image, as follows:

$$x'_0 = |F^{-1}(f_l + f_h)|, \quad (12)$$

where x'_0 is the reorganized image of frequency domain.

After that, we can obtain the next iterative image x'_{t-1} of frequency domain, and we introduce residual connections to further suppress the influence of artifacts during the reverse process,

$$x'_{t-1} = \kappa \cdot \left(\frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x'_0 \right) + (1 - \kappa) \cdot x'_0, \quad (13)$$

where $\kappa = 2 - 2^{(1-\frac{t}{T})}$, which can effectively suppress artifact interference and correct data flow during the reverse process.

3.2.3. Pixel domain removal

Only focusing on the restoration of high-frequency details while ignoring low-frequency information will result in a lack of naturalness in the overall image. In order to enhance the naturalness of the image, we perform motion artifact removal in the pixel domain. Pixel domain removal can also handle the slight phase loss caused by the modulus operation in frequency domain removal and further standardize data flow during the reverse process. The algorithm of this module is shown in Algorithm 2.

We reorganize $\hat{x}_0$ using alternate complementary masks in the same way as frequency domain removal. The expression is as follows:

$$x''_0 = x_m \odot \mathcal{M}_t + \hat{x}_0 \odot (1 - \mathcal{M}_t), \quad (14)$$

where x''_0 is the reorganised image of pixel domain. Then, we can obtain the next iterative x''_{t-1} of pixel domain,

$$x''_{t-1} = \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x''_0, \quad (15)$$

Algorithm 2 Pixel domain removal.

Input: Current image x_t , motion-corrupted image x_m , the predicted image $\hat{x}_0$, intrinsic variable of diffusion model $\bar{\alpha}_t$, the alternate complementary masks $\mathcal{M}_t$.

- 1: Combine x_m and x_t through $\mathcal{M}_t$ to obtain the reorganized image x''_0 using Eq. (14);
- 2: Obtain the next iterative image x''_{t-1} of pixel domain using Eq. (15);

Output: Updated image in pixel domain x''_{t-1} .

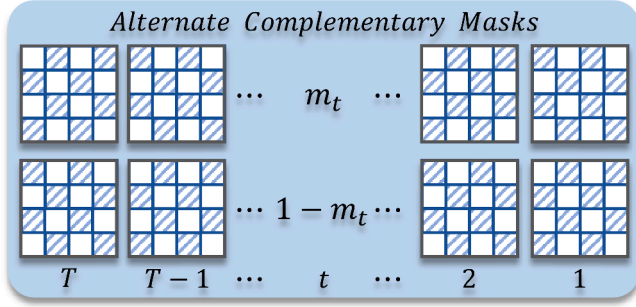


Fig. 3. The illustration of alternate complementary masks. In each iteration, the complementary masks are alternately transformed to ensure artifact removal for the entire image.

3.2.4. Alternate complementary masks

Using masks can destroy the artifact distribution while retaining part of the useful information. In previous research, the mask used in diffusion models was fixed (Lugmayr et al., 2022). However, in our method, masks are flipped at each reverse step to alternatively cover all image areas throughout the process, as shown in Fig. 3, which is expressed as follows:

$$m_{t-1} = 1 - m_t. \quad (16)$$

During the reverse process, although the alternate complementary masks destroy artifact distribution, the motion artifact information retained by the mask still seriously affect the diffusion model in later stages. Therefore, we introduce a variable to fine-tune the weighting of the masks, aiming to alleviate this influence.

$$\mathcal{M}_t = \omega_t \cdot m_t, \quad (17)$$

where $\omega_t = 1 - \sqrt{\bar{\alpha}_t}$ which is an intrinsic variable of DDPM.

In (17), as time t gradually decreases from T , ω_t converges from 1 to 0. This shows that in the initial stage of the reverse process, the model gradually generates the correct shape and textures by useful information. Additionally, as the reorganised image still contains a amount of Gaussian noise at this stage, the influence of motion artifacts can be disregarded. In the later stages, motion artifacts can seriously interfere with the diffusion model. Considering that the generated image already has correct shape and textures at this stage, the model is no longer completely dependent on the guidance of the useful information, so we reduce the value of ω_t to mitigate the interference of motion artifacts.

3.2.5. Dual domain balance

In order to remove motion artifacts while ensuring visual clarity and naturalness of the images, we introduce a parameter γ_t that varies with the diffusion step t ,

$$\gamma_t = -a \cdot e^{-\frac{t}{T}} + 1, \quad (18)$$

where a is a tunable hyperparameter ranging from 0 to 1. γ_t is equal to $1 - \frac{a}{e}$ when $t = T$. As t gradually decreases, γ_t gradually approaches $1 - a$.

We use γ_t to balance the recovered images between the frequency domain and pixel domain, as follows:

$$\tilde{x}_{t-1} = \gamma_t \cdot x'_{t-1} + (1 - \gamma_t) \cdot x''_{t-1}, \quad (19)$$

where $\tilde{x}_{t-1}$ is the final reorganised image.

According to the definition of γ_t , in the initial stages of the reverse process, we focus more on recovering information in frequency domain to make the reorganised image contain more tissue textures. As t gradually decreases, we gradually increase the proportion of information from the pixel domain, aiming to achieve a dual objective of balancing artifact removal and visual clarity.

Algorithm 3 MAPPK for motion artifact removal.

Input: The motion-corrupted image x_m , the diffusion step $\hat{t}$, high-pass and low-pass filters Φ_h, Φ_l , the diffusion model D_θ , the intrinsic variable sequence of diffusion model $\{\bar{\alpha}_t\}_1^{\hat{t}}$, mask sequence $\{m_j\}_1^{\hat{t}}$, prior meanbook $\{\bar{x}_s\}_1^k$, trade-off hyperparameter η , dual-domain balance hyperparameter a .

- 1: Calculate the image x_i using Eq. (20);
- 2: **for** $t = \hat{t}$ **to** 1 **do**
- 3: Input x_t into D_θ to get the predicted image $\hat{x}_0$;
 $\hat{x}_0 \leftarrow D_\theta(x_t)$;
- 4: Calculate the alternate complementary masks $\mathcal{M}_t$ using Eq. (17);
- 5: Obtain the next iterative image x'_{t-1} of frequency domain by executing Algorithm 1;
- 6: Obtain the next iterative image x''_{t-1} of pixel domain by executing Algorithm 2;
- 7: Calculate the dual domain balanced parameter γ_t using Eq. (18);
- 8: Obtain the balanced images $\tilde{x}_{t-1}$ between frequency domain and pixel domain using Eq. (19);
- 9: $x_{t-1} \leftarrow \tilde{x}_{t-1}$;
- 10: **end for**

Output: The clean image x_0 .

3.2.6. Accelerate inference

Since we set the total number of timesteps T to 1000, initiating the reverse process from x_T is time-consuming, and an excessively long timestep introduces greater uncertainty into the reverse process. Moreover, during the forward process, as t increases, motion artifacts are progressively overwhelmed by Gaussian noise, leading to a diminishing difference between clean and motion-corrupted images. Therefore, starting inference from x_T becomes unnecessary and we choose to initiate the reverse process from an intermediate timestep to improve efficiency.

Specifically, we add noise to the motion-corrupted image x_m to level $\hat{t}$,

$$x_i = \sqrt{\bar{\alpha}_i} \cdot x_m + \sqrt{1 - \bar{\alpha}_i} \cdot \epsilon. \quad (20)$$

Then, we initiate the inference from x_i to improve the efficiency and reduce the uncertainty of the reverse process. Based on empirical observations, we set $\hat{t} = 200$. The details of the overall process are presented in Algorithm 3.

4. Implementation details

4.1. Experimental datasets

We utilize three MRI datasets for evaluating and validating the proposed MAPPK method.

4.1.1. Brain

The first dataset is the Human Connectome Project (HCP) data, which is a publicly available dataset containing MRI data of the human brain. This dataset is obtained by a Siemens 3T system with a 3D spin-echo sequence. The imaging parameters for acquiring the HCP data are as follows: TR = 3200 ms, TE = 565 ms, voxel size = 0.7 mm × 0.7 mm × 0.7 mm, matrix size = 320 × 320, number of slices = 256, echo train duration = 1105, echo spacing = 3.53 ms, and the phase encoding direction = anterior-posterior. We selected 3000 motion-free images from 150 subjects with 20 images per subject for training the diffusion model. Additionally, 800 images were chosen from 40 subjects with 20 images per subject for generating motion-corrupted images and evaluation. All images are center-cropped to the size of 256 × 256 to remove redundant areas that do not contain brain information.

4.1.2. Knee

The second dataset is fastMRI multi-coil data (Zbontar et al., 2018). We use the knee MRI data from this dataset. The number of coils of this data is 15. The acquisition parameters are as follows: TR = 2200 - 3000 ms, TE = 27 - 34 ms, matrix size = 320 × 320, in-plane resolution

= 0.5 mm × 0.5 mm, slice thickness = 3 mm, and phase encoding direction = anterior-posterior. We selected 3000 motion-free images for training and 800 images for generating motion-corrupted images and evaluation.

4.1.3. Abdominal

The third dataset is the T2-weighted abdominal images from Department of Radiology, Xiangyang No. 1 People's Hospital, Hubei University of Medicine. This study was approved by the local ethics committee, and the requirement for informed consent was waived because of the retrospective and anonymous analysis of the data (XYYYE20240082). All clinical data are anonymized before being used in this study. The dataset is obtained by a 3T Philips Ingenia MR system with the following scan parameters: TR = 2130.6 ms, TE = 80 ms, slice thickness/intersection gap = 5/0 mm, echo time = 80 ms, field of view = 100 mm × 100 mm, acquisition matrix = 288 × 288, the phase encoding direction = anterior-posterior. After acquisition, zero padding was applied to the acquired k -space, resulting in image sizes of 640 × 640 after padding. All images are center-cropped to the size of 512 × 512. We selected 3002 motion-free images from 94 subjects for training and 576 images from 18 subjects to generate motion-corrupted images and evaluate the performance of our method. Additionally, 40 real motion-corrupted images from clinical settings are used for qualitative and radiologist evaluations.

4.2. Simulation of motion artifact

Similar to previous works (Oh et al., 2021; Tamada et al., 2020), the simulation of motion artifacts can be expressed as follows,

$$\hat{\mathbf{x}}(k_x, k_y) = \begin{cases} \mathbf{x}(k_x, k_y)e^{-j\Phi(k_y)}, & k_y \in \mathbb{K} \\ \mathbf{x}(k_x, k_y), & \text{otherwise,} \end{cases} \quad (21)$$

where $\hat{\mathbf{x}}$ and $\mathbf{x}$ refer to the motion-corrupted and motion-free k -space data, $j = \sqrt{-1}$, k_x and k_y are the indices along the read-out and phase encoding directions, respectively. $\Phi(k_y)$ is the displacement in radian along the phase encoding direction, and $\mathbb{K}$ is the position in the phase-encoding direction where the perturbations occur.

For brain and knee datasets, we simulate random motion artifacts of translation and rotation. $\Phi(k_y)$ in Eq. (21) can be represented as follows,

$$\Phi(k_y) = \begin{cases} \mathcal{R}\{k_y\Delta_k\}, & |k_y| > k_0 \\ 0, & \text{otherwise,} \end{cases} \quad (22)$$

where Δ_k denotes the degree of motion along y direction, $\mathcal{R}$ denotes the rotation angle of the image in pixel domain, and k_0 is the delay time of the phase perturbation (Tamada et al., 2020). In this simulation, k_0 is fixed to $\pi/10$ and $\pi/20$, representing mild and severe motion artifacts, respectively, Δ_k is adjusted to be 2.5–3.0 cm, and $\mathcal{R}$ is sampled from $[-2^\circ, 2^\circ]$.

For abdominal dataset, we simulate respiratory motion artifacts as follows,

$$\Phi(k_y) = \begin{cases} k_y\Delta_k \sin(mk_y + n), & |k_y| > k_0 \\ 0, & \text{otherwise,} \end{cases} \quad (23)$$

where Δ_k , m , and n denote the amplitude, period, and phase perturbation, respectively. In this simulation, Δ_k is adjusted to be 1.1–1.2 cm, m is randomly sampled from $[0.1, 5.0]$, and n is randomly sampled from $[0, \pi/4]$.

4.3. Comparison approaches

We select three classes of approaches for comparison. The first class is supervised approach, Pix2pix (Isola et al., 2017), which uses paired images for model training. The second class is unsupervised approaches, CycleGAN (Zhu et al., 2017) and UDDN (Wu et al., 2023), both of which can be trained with either simulated or real motion-corrupted images.

Table 1

Partial parameter setting of the diffusion model during training and inference.

Size	Dim	Mult-Ch	N-Res	Head-Ch	N-Head
256	128	{1, 1, 2, 2, 4, 4}	2	64	4
320	128	{1, 1, 2, 4, 4, 4}	2	64	4
512	128	{0.5, 1, 1, 2, 2, 4, 4}	2	64	4

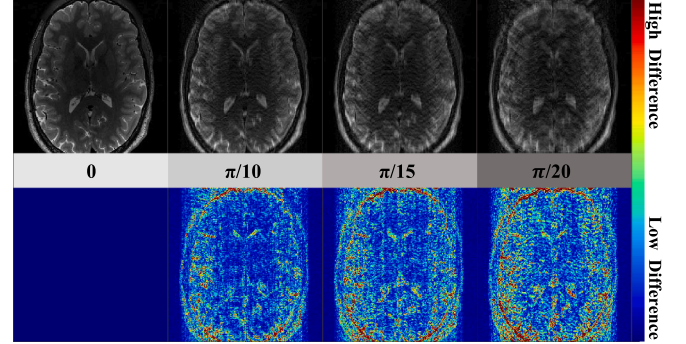


Fig. 4. Visual comparison of motion-corrupted brain images at different cut-off frequencies (0, $\pi/10$, $\pi/15$, and $\pi/20$). The first row shows the brain images, while the last row shows the corresponding difference heatmaps compared to the ground truth (0). The color ranges from blue to red, indicating differences from small to large, with deeper colors representing smaller differences. The image ceases to exhibit clear textural structures with the cut-off frequency of $\pi/20$. Taking into account the timing of patient motion in clinical scenarios, the effective cut-off frequency is typically above $\pi/20$.

However, training with real images often leads to unstable training (Oh et al., 2023), so we only use simulated motion-corrupted images for training. The third class is unsupervised approaches based on diffusion model. We use DR2 (Wang et al., 2023), GDP (Fei et al., 2023), and RDBM (Wang et al., 2026) for comparison, all of which are novel approaches for image recovery. We also compare our method with PFAD (Xu et al., 2025) to verify its effectiveness of the improvement, noting that the cut-off value of PFAD also varies with the experimental setting. The training and validation parameters of these approaches are set according to the parameters specified in the original papers to achieve optimal motion artifact removal effects, respectively.

4.4. Training and evaluation details

4.4.1. Training

We use guided diffusion (Dhariwal & Nichol, 2021) for training. Some important parameters in the training process are listed in Table 1, where Size is the size of the input image, Dim is the number of the channel dimensions, Mult-Ch denotes the channel multipliers, N-Res denotes the number of residual block, Head-Ch denotes the number of channels per attention head, and N-Head is the number of heads in the Multi-Head attention mechanism. We set the batch size to 4, the total number of timesteps to 1000, and learning rate to 1.0×10^{-4} . We trained the model with 50k, 200k, and 500k iterations respectively on size 256, 320, and 512. The diffusion model training is conducted on two RTX 4090 GPUs. We used the AdamW optimizer with default parameters, where the beta parameters are set to (0.9, 0.999).

4.4.2. Evaluation

We use five metrics to quantitatively evaluate the effectiveness on the simulated images: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) (Wang et al., 2004), root mean square error (RMSE), Relative Root Mean Square Error (RRMSE), feature similarity index measure (FSIM) (Zhang et al., 2011), visual information

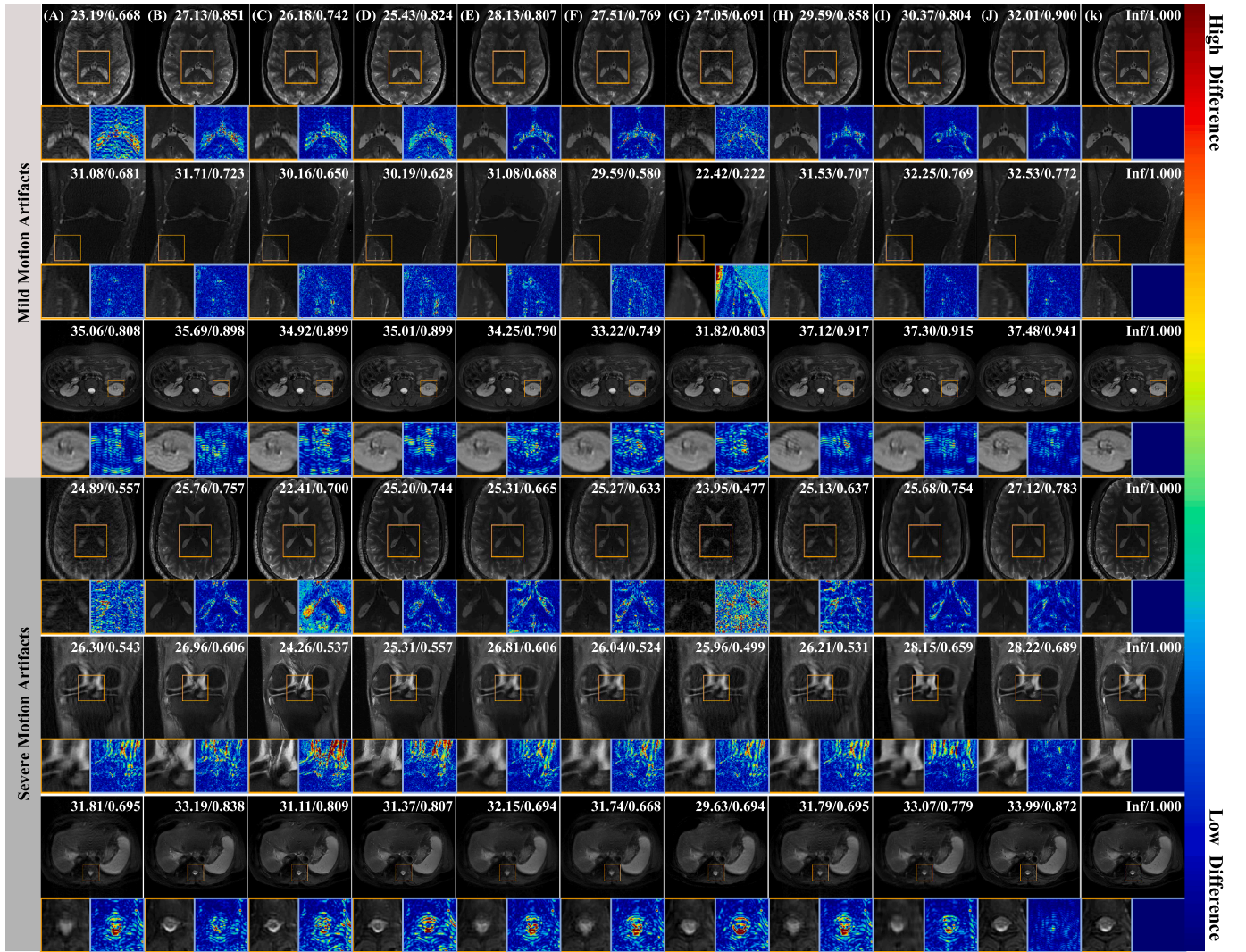


Fig. 5. Motion artifact removal results using various methods with simulated random motion artifacts: (A) Artifact images, (B) Pix2pix, (C) CycleGAN, (D) UDDN, (E) DR2, (F) GDP, (G) BIRD (H) PFAD, (I) RDBM, (J) MAPPK (ours), and (K) the ground-truth. PSNR and SSIM values of each image are shown in the corner of images. The top rows show the comparison results of mild motion artifacts, and the bottom rows show the comparison results of severe artifacts. The yellow box indicates the zoomed-in visualization area, and the blue box represents the difference heatmap compared to the ground truth. The color ranges from blue to red, indicating differences from small to large, with deeper colors representing smaller differences. We use global normalization for fair visual comparison of error magnitudes.

fidelity (VIF) (Sheikh & Bovik, 2006), visual saliency-induced Index (VSI) (Zhang et al., 2014) and gradient magnitude similarity deviation (GMSD) (Xue et al., 2013). Meanwhile, we perform a qualitative evaluation on the clinical images. Two young radiologists with 3 years of diagnostic experience independently evaluated the denoising results without knowing any clinical details or biopsy results, using a Likert 5-scale (Likert, 1932) scoring system where higher scores indicate better performance. The evaluation metrics included artifact removal performance, image blurriness, anatomical deformation, diseased tissue deformation, and overall image quality. If the opinions of the two young radiologists were consistent, their scores were used as the final scores. Otherwise, another radiologist with 20 years of experience conducted a further review and made the final decision. For statistical analysis, the Mann-Whitney U test was performed on the image-level quantitative results of the entire test set. The sample sizes were 800 images for the brain dataset, 800 images for the knee dataset, and 576 images for the abdominal dataset. For each individual metric, the proposed method was independently compared with each baseline method on the same test set. Thus, each reported p -value represents the statistical difference between one specific baseline and our method under one specific metric.

To address the complexity of artifacts in clinical settings, we introduced an adaptive selection mechanism based on different cut-off frequencies during clinical evaluation. Specifically, we first generate simulated artifact images at different cut-off frequencies using Eqs. (22) and (23). We then constructed a K -Nearest Neighbors (KNN) classification model utilising their statistical characteristics (mean, standard deviation, and grey-level histogram, among others) to distinguish between different artifact categories. In the inference process, the KNN model predicts the artifact category based on the statistical features of the input image, thereby adaptively selecting the most appropriate cut-off frequency. All approaches requiring additional training at different cut-off frequencies were trained accordingly. It should be noted that the cut-off frequency adjustment range was set between $\pi/10$ and $\pi/20$, with multiple steps at reasonable intervals (such as $\pi/10$, $\pi/15$ and $\pi/20$). The selection of this range takes into account both clinical applicability (Oh et al., 2023) and the visual resolution ability of radiologists (as shown in Fig. 4). Compared to approaches requiring separate model training for each cut-off frequency, our method offers greater flexibility without additional training. Furthermore, relative to other universal artifact removal methods that do not require training, our method adapts to

Table 2

Quantitative comparison of different methods on simulated images with the cut-off frequency of $\pi/10$. The values in the table are reported as mean values with the half-width of the 95% confidence interval for the entire test dataset. **Bold** represents the best and underlined represents the second best. All results underwent Mann-Whitney U test, (*) indicates $p \leq 0.05$, and (**) indicates $p \leq 0.01$.

Datasets	Methods	Year	PSNR (dB) $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$	RRMSE $\downarrow$	FSIM $\uparrow$	VIF $\uparrow$	VSI $\uparrow$	GMSD $\downarrow$
Brain Random	Input	—	25.97** ± 1.06	0.685** ± 0.029	0.051** ± 0.006	0.253** ± 0.020	0.812** ± 0.016	0.295** ± 0.022	0.945** ± 0.007	0.114** ± 0.014
	Pix2pix	2017	26.76** ± 1.08	<u>0.841**</u> ± 0.015	0.046** ± 0.006	0.232** ± 0.026	0.882** ± 0.011	0.392** ± 0.019	0.966** ± 0.004	0.065** ± 0.007
	CycleGAN	2017	25.94** ± 1.00	0.719** ± 0.025	0.051** ± 0.006	0.256** ± 0.039	0.846** ± 0.013	0.338** ± 0.020	0.955** ± 0.005	0.091** ± 0.009
	UDDN	2023	26.07** ± 1.03	0.823** ± 0.018	0.050** ± 0.006	0.251** ± 0.036	0.879** ± 0.010	0.372** ± 0.019	0.965** ± 0.004	0.065** ± 0.006
	DR2	2023	26.60** ± 1.06	0.780** ± 0.024	0.047** ± 0.006	0.235** ± 0.018	0.859** ± 0.013	0.372** ± 0.024	0.960** ± 0.005	0.083** ± 0.009
	GDP	2023	26.08** ± 0.96	0.739** ± 0.022	0.050** ± 0.006	0.250** ± 0.018	0.847** ± 0.011	0.313** ± 0.017	0.957** ± 0.005	0.092** ± 0.008
	BIRD	2024	25.62** ± 0.96	0.670** ± 0.037	0.053** ± 0.006	0.263** ± 0.021	0.812** ± 0.014	0.269** ± 0.021	0.944** ± 0.007	0.110** ± 0.012
	PFAD	2025	27.60** ± 1.13	0.829** ± 0.020	0.042** ± 0.005	0.210** ± 0.018	0.896** ± 0.012	0.426** ± 0.024	0.970** ± 0.004	0.061** ± 0.009
	RDBM	2026	28.66** ± 1.14	0.787** ± 0.033	0.037** ± 0.005	0.186** ± 0.015	0.903** ± 0.012	0.475** ± 0.026	0.972** ± 0.004	0.055 ± 0.007
	MAPPK (ours)	—	29.22 ± 1.19	0.868 ± 0.020	0.035 ± 0.005	0.174 ± 0.015	0.913 ± 0.011	0.502 ± 0.028	0.975 ± 0.004	0.055 ± 0.009
Knee Random	Input	—	29.05** ± 1.77	0.660** ± 0.066	0.036** ± 0.007	0.158** ± 0.039	0.846** ± 0.022	0.266** ± 0.052	0.965** ± 0.008	0.110** ± 0.023
	Pix2pix	2017	29.37** ± 1.55	0.705** ± 0.063	0.035** ± 0.006	0.152** ± 0.035	0.861** ± 0.018	0.296** ± 0.051	0.970** ± 0.005	0.085** ± 0.014
	CycleGAN	2017	27.34** ± 2.02	0.705** ± 0.089	0.044** ± 0.011	0.197** ± 0.071	0.849** ± 0.021	0.276** ± 0.072	0.967** ± 0.006	0.095** ± 0.015
	UDDN	2023	28.50** ± 1.56	0.663** ± 0.090	0.038** ± 0.007	0.170** ± 0.047	0.852** ± 0.021	0.274** ± 0.072	0.967** ± 0.006	0.089** ± 0.015
	DR2	2023	29.23** ± 1.61	0.653** ± 0.083	0.035** ± 0.006	0.155** ± 0.039	0.835** ± 0.026	0.300** ± 0.064	0.964** ± 0.007	0.101** ± 0.014
	GDP	2023	28.08** ± 1.53	0.599** ± 0.089	0.040** ± 0.007	0.177** ± 0.046	0.840** ± 0.018	0.213** ± 0.058	0.966** ± 0.006	0.104** ± 0.017
	BIRD	2024	28.19** ± 2.19	0.603** ± 0.111	0.040** ± 0.013	0.188** ± 0.120	0.836** ± 0.037	0.278** ± 0.087	0.964** ± 0.011	0.114** ± 0.026
	PFAD	2025	29.64** ± 1.63	0.712** ± 0.071	0.034** ± 0.006	0.148** ± 0.037	0.869** ± 0.018	0.314** ± 0.065	0.972** ± 0.005	0.080 ± 0.016
	RDBM	2026	30.49** ± 1.63	0.764** ± 0.061	0.030** ± 0.006	0.134** ± 0.034	0.869** ± 0.022	0.386** ± 0.060	0.972** ± 0.006	0.080** ± 0.015
	MAPPK (ours)	—	30.70 ± 1.73	0.772 ± 0.060	0.030 ± 0.006	0.131 ± 0.033	0.869** ± 0.022	0.386** ± 0.054	0.972** ± 0.006	0.080** ± 0.017
Abdominal Respiratory	Input	—	35.37** ± 1.51	0.816** ± 0.037	0.017** ± 0.003	0.123** ± 0.020	0.944** ± 0.010	0.441** ± 0.043	0.986** ± 0.003	0.064** ± 0.011
	Pix2pix	2017	36.14** ± 1.37	0.902** ± 0.020	0.016** ± 0.003	0.112** ± 0.016	0.955** ± 0.008	0.481** ± 0.030	0.988** ± 0.002	0.058** ± 0.009
	CycleGAN	2017	34.68** ± 1.91	0.902** ± 0.024	0.019** ± 0.005	0.135** ± 0.039	0.949** ± 0.012	0.468** ± 0.046	0.986** ± 0.003	0.067** ± 0.012
	UDDN	2023	35.14** ± 1.42	0.896** ± 0.027	0.018** ± 0.003	0.126** ± 0.020	0.951** ± 0.010	0.473** ± 0.038	0.965** ± 0.003	0.066** ± 0.010
	DR2	2023	34.37** ± 1.19	0.794** ± 0.033	0.019** ± 0.003	0.137** ± 0.020	0.933** ± 0.010	0.389** ± 0.024	0.983** ± 0.003	0.076** ± 0.008
	GDP	2023	33.41** ± 1.05	0.760** ± 0.032	0.022** ± 0.003	0.153** ± 0.023	0.919** ± 0.012	0.320** ± 0.022	0.981** ± 0.003	0.083** ± 0.008
	BIRD	2024	29.31** ± 3.63	0.725** ± 0.127	0.038** ± 0.019	0.288** ± 0.197	0.867** ± 0.058	0.271** ± 0.076	0.968** ± 0.022	0.120** ± 0.039
	PFAD	2025	37.24** ± 1.08	0.918** ± 0.009	0.014** ± 0.002	0.099** ± 0.014	0.962** ± 0.005	0.544** ± 0.025	0.990** ± 0.002	0.051** ± 0.006
	RDBM	2026	37.23** ± 1.41	0.910** ± 0.015	0.014** ± 0.002	0.099** ± 0.016	0.961** ± 0.008	0.562** ± 0.043	0.990** ± 0.002	0.052** ± 0.009
	MAPPK (ours)	—	37.46 ± 1.30	0.942 ± 0.013	0.014 ± 0.002	0.096 ± 0.015	0.968 ± 0.007	0.585 ± 0.031	0.991 ± 0.002	0.048 ± 0.007

varying cut-off frequencies, demonstrating superior generalisation capabilities when confronting complex clinical artifacts. This enables more precise and reliable image restoration.

5. Experimental results

5.1. Quantitative evaluation of simulated images

Fig. 5 shows the visual comparison of different methods on simulated images. The input image (A) clearly shows motion-induced structural distortion and streaky artifacts. Pix2pix (B) introduces local deformations and generates inaccurate texture structures. CycleGAN (C) is unable to effectively remove motion artifacts and often produces blur, especially in areas with subtle tissue contrast. UDDN (D) has limited preservation of image structure and generates spurious textures. DR2 (E) tends to over-smooth fine brain textures, removing key anatomical details, while GDP (F) misinterprets artifact-related intensity changes as structural information, resulting in visually inconsistent tissue representations. BIRD (G) is unable to remove artifacts and produces a large amount of noise, reflecting its limited ability to handle complex motion-induced artifacts. PFAD (H), combined with the pixel-frequency information, demonstrates good restoration results for mild artifacts; however, its reconstruction ability deteriorates significantly when faced with severe artifacts. This is because the low-frequency information on which it is based contains artifact information, and the model cannot recover the accurate texture structure. RDBM (I) improves the restoration quality compared with several previous methods, but residual artifacts and local texture inconsistency can still be observed. In contrast, MAPPK (J) effectively removes artifacts and maintains global structural integrity and local texture details.

For the knee simulation images, the input image (A) exhibits noticeable stripe-like motion artifacts and localized structural blurring caused

by simulated motion, particularly around the soft tissue and bone edges. Pix2pix (B) shows limited performance in artifact removal—the generated results still contain considerable residual artifacts and inaccurate texture reconstruction, leading to structural discontinuities and blurring; its synthesis quality drops significantly under severe artifacts. CycleGAN (C) can partially suppress motion artifacts but produces unstable brightness and contrast, with over-enhancement and texture distortion in some regions, resulting in incomplete artifact removal. UDDN (D) fails to recover high-frequency edge details, producing blurred textures and losing fine structural information. The results of DR2 (E) are overly smooth and the most high-frequency information is lost, weakening the fine structures. GDP (F) misinterprets motion artifacts as structural textures, leading to substantial artifact retention. Similarly, BIRD (G) performs poorly under severe motion artifacts, with generated images showing contrast deviations from real anatomy and local structural distortions. PFAD (H) achieves some improvement in structural texture recovery, but residual artifacts remain visible under severe motion conditions. RDBM (I) produces more stable restoration results than most comparison methods, but some residual artifacts and texture inaccuracies remain. In comparison, MAPPK (J) demonstrates clear superiority, effectively removing motion artifacts while preserving fine-grained tissue textures and clear structural boundaries. The restored images exhibit realistic anatomical contrast, clear soft tissue layers, and sharp bone edges, which highlights the strong adaptability and excellent reconstruction capability of MAPPK in complex motion artifact scenarios.

For the abdominal simulation images, the input image (A) shows noticeable blurring of texture structures caused by motion. Pix2pix (B) fails to effectively remove motion artifacts, resulting in obvious residual artifacts. CycleGAN (C) exhibits limited artifact removal performance, with local brightness mismatches and loss of fine structures. UDDN (D) suffers a significant performance decline under severe artifacts, failing to reconstruct fine structural texture details. DR2 (E) produces overly

Table 3

Quantitative comparison of different methods on simulated images with the cut-off frequency of $\pi/20$. The values in the table are reported as mean values with the half-width of the 95% confidence interval for the entire test dataset. **Bold** represents the best and underlined represents the second best. All results underwent Mann-Whitney U test, (*) indicates $p \leq 0.05$, and (**) indicates $p \leq 0.01$.

Datasets	Methods	Year	PSNR (dB) $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$	RRMSE $\downarrow$	FSIM $\uparrow$	VIF $\uparrow$	VSI $\uparrow$	GMSD $\downarrow$
Brain Random	Input	—	22.93** ± 1.06	0.524** ± 0.036	0.072** ± 0.009	0.359** ± 0.029	0.737** ± 0.019	0.158** ± 0.019	0.922** ± 0.009	0.177** ± 0.015
	Pix2pix	2017	24.11** ± 1.14	0.720** ± 0.024	0.063** ± 0.008	0.313** ± 0.020	<u>0.818**</u> ± 0.014	0.242** ± 0.021	<u>0.944**</u> ± 0.007	0.122 ± 0.010
	CycleGAN	2017	23.46** ± 1.03	0.707** ± 0.029	0.068** ± 0.008	0.339** ± 0.040	0.813** ± 0.016	0.233** ± 0.022	0.941** ± 0.007	0.129** ± 0.010
	UDDN	2023	23.63** ± 1.07	0.703** ± 0.031	0.066** ± 0.008	0.331** ± 0.023	0.810** ± 0.016	0.231** ± 0.023	0.940** ± 0.007	<u>0.129**</u> ± 0.010
	DR2	2023	23.41** ± 1.07	0.617** ± 0.038	0.068** ± 0.008	0.340** ± 0.027	0.783** ± 0.018	0.207** ± 0.025	0.936** ± 0.007	0.148** ± 0.012
	GDP	2023	23.36** ± 1.06	0.589** ± 0.039	0.068** ± 0.008	0.342** ± 0.027	0.777** ± 0.019	0.194** ± 0.022	0.936** ± 0.007	0.150** ± 0.013
	BIRD	2024	22.66** ± 0.97	0.498** ± 0.038	0.074** ± 0.008	0.371** ± 0.028	0.736** ± 0.016	0.145** ± 0.017	0.918** ± 0.008	0.173** ± 0.013
	PFAD	2025	23.07** ± 1.08	0.584** ± 0.038	0.071** ± 0.009	0.353** ± 0.030	0.772** ± 0.020	0.175** ± 0.022	0.934** ± 0.008	0.163** ± 0.015
	RDBM	2026	25.10** ± 1.16	<u>0.724**</u> ± 0.029	<u>0.056</u> ± 0.008	<u>0.280**</u> ± 0.022	0.811** ± 0.018	0.298** ± 0.026	0.942** ± 0.008	0.133** ± 0.015
	MAPPK (ours)	—	25.19 ± 1.15	0.735 ± 0.032	0.055 ± 0.007	0.277 ± 0.023	0.830 ± 0.020	0.316 ± 0.031	0.948 ± 0.008	0.122 ± 0.014
Knee Random	Input	—	25.79** ± 1.98	0.512** ± 0.071	0.053** ± 0.012	0.228** ± 0.046	0.785** ± 0.027	0.126** ± 0.031	0.951** ± 0.011	0.169** ± 0.029
	Pix2pix	2017	26.61** ± 1.59	0.576** ± 0.062	0.048** ± 0.009	0.207** ± 0.043	<u>0.810**</u> ± 0.019	0.168** ± 0.040	<u>0.955**</u> ± 0.007	<u>0.145**</u> ± 0.018
	CycleGAN	2017	24.44** ± 1.80	0.541** ± 0.088	0.061** ± 0.013	0.271** ± 0.078	0.788** ± 0.024	0.146** ± 0.047	0.951** ± 0.008	0.154** ± 0.020
	UDDN	2023	25.95** ± 1.76	0.563** ± 0.084	0.051** ± 0.010	0.225** ± 0.049	0.798** ± 0.024	0.158** ± 0.046	0.956 ± 0.008	0.147** ± 0.022
	DR2	2023	26.25** ± 1.98	0.564** ± 0.080	0.050** ± 0.011	0.216** ± 0.043	0.788** ± 0.023	0.158** ± 0.038	<u>0.955**</u> ± 0.008	0.155** ± 0.025
	GDP	2023	25.59** ± 1.83	0.491** ± 0.077	0.054** ± 0.011	0.233** ± 0.046	0.787** ± 0.024	0.120** ± 0.033	0.954** ± 0.009	0.160** ± 0.028
	BIRD	2024	25.05** ± 2.00	0.466** ± 0.084	0.057** ± 0.014	0.258** ± 0.111	0.775** ± 0.032	0.132** ± 0.067	0.949** ± 0.012	0.175** ± 0.028
	PFAD	2025	25.70** ± 1.95	0.497** ± 0.078	0.053** ± 0.012	0.230** ± 0.046	0.782** ± 0.026	0.122** ± 0.031	0.951** ± 0.010	0.169** ± 0.028
	RDBM	2026	26.97** ± 1.70	<u>0.626**</u> ± 0.064	<u>0.046</u> ± 0.009	<u>0.198</u> ± 0.039	0.800** ± 0.024	<u>0.213**</u> ± 0.044	0.956 ± 0.009	0.146** ± 0.020
	MAPPK (ours)	—	27.13 ± 2.03	0.643 ± 0.062	0.045 ± 0.010	0.195 ± 0.038	0.819 ± 0.028	0.222 ± 0.032	0.956 ± 0.009	0.144 ± 0.024
Abdominal Respiratory	Input	—	30.56** ± 1.43	0.651** ± 0.055	0.030** ± 0.005	0.213** ± 0.034	0.868** ± 0.018	0.209** ± 0.035	0.967** ± 0.006	0.113** ± 0.014
	Pix2pix	2017	31.19** ± 1.39	<u>0.820**</u> ± 0.031	0.028** ± 0.005	0.197** ± 0.026	<u>0.893**</u> ± 0.017	0.257** ± 0.024	0.969** ± 0.005	0.111** ± 0.012
	CycleGAN	2017	30.02** ± 1.55	0.808** ± 0.040	0.032** ± 0.006	0.228** ± 0.049	0.883** ± 0.022	0.248** ± 0.037	0.967** ± 0.006	0.118** ± 0.013
	UDDN	2023	30.62** ± 1.28	0.815** ± 0.036	0.030** ± 0.005	0.211** ± 0.032	0.889** ± 0.021	0.252** ± 0.032	0.968** ± 0.005	0.116** ± 0.013
	DR2	2023	30.76** ± 1.28	0.642** ± 0.050	0.029** ± 0.004	0.208** ± 0.030	0.880** ± 0.018	0.240** ± 0.027	0.969** ± 0.005	0.115** ± 0.011
	GDP	2023	30.56** ± 1.24	0.623** ± 0.048	0.030** ± 0.004	0.213** ± 0.031	0.877** ± 0.019	0.217** ± 0.024	0.968** ± 0.005	0.115** ± 0.012
	BIRD	2024	27.59** ± 2.71	0.677** ± 0.108	0.044** ± 0.016	0.329** ± 0.178	0.836** ± 0.046	0.177** ± 0.040	0.955** ± 0.018	0.142** ± 0.028
	PFAD	2025	30.64** ± 1.38	0.655** ± 0.050	0.030** ± 0.005	0.211** ± 0.032	0.878** ± 0.018	0.234** ± 0.037	0.969** ± 0.005	0.110** ± 0.013
	RDBM	2026	31.63** ± 1.33	0.720** ± 0.048	0.027** ± 0.004	0.188** ± 0.025	<u>0.893**</u> ± 0.019	<u>0.325**</u> ± 0.036	0.971** ± 0.005	0.110** ± 0.013
	MAPPK (ours)	—	32.82 ± 1.24	0.869 ± 0.026	0.023 ± 0.003	0.164 ± 0.024	0.917 ± 0.016	0.356 ± 0.027	0.977 ± 0.004	0.093 ± 0.011

Table 4

Evaluated by radiologists using Likert 5-scale scoring system. Higher scores indicate higher performance. The results are presented as mean/variance.

Methods	Artifacts	Blurring	Anatomical	Diseased	Overall
Pix2pix	3.85/0.89	3.88/0.79	4.35/0.70	4.35/0.74	3.75/0.81
CycleGAN	3.13/0.94	3.85/0.83	4.30/0.79	4.28/0.78	3.33/0.80
UDDN	3.50/1.18	3.93/0.86	4.30/0.72	4.20/0.79	3.58/0.90
DR2	4.30/0.82	4.00/0.75	4.40/0.78	4.38/0.74	4.13/0.72
GDP	4.25/0.81	3.98/0.80	4.28/0.78	4.25/0.87	3.90/0.78
BIRD	2.78/1.25	2.58/1.15	2.03/1.33	2.18/1.08	1.98/1.10
PFAD	4.35/0.80	3.98/0.73	4.38/0.77	4.45/0.60	4.15/0.74
RDBM	4.47/0.42	4.25/0.61	4.39/0.56	4.37/0.75	4.02/0.55
MAPPK	4.83/0.38	4.45/0.68	4.75/0.49	4.80/0.46	4.65/0.48

smooth recovery, unable to maintain realistic textures and structural consistency. GDP (F) still misinterprets motion artifacts as texture, leaving a large amount of residual artifacts and failing to recover accurate structural details under severe motion conditions. BIRD (G) can partially remove artifacts but generates inaccurate texture structures with visible artifact traces. PFAD (H) introduces additional stripe-like artifacts during generation and cannot effectively eliminate artifacts under severe motion interference. RDBM (I) improves artifact suppression to some extent, but local texture inconsistency and residual artifacts are still visible. In contrast, MAPPK (J) demonstrates clear superiority in handling respiratory motion artifacts. It effectively removes complex motion-induced distortions while accurately restoring the anatomical structures and fine textures. The generated images exhibit natural contrast, clear structural layers, and superior overall visual quality compared to other approaches.

Tables 2 and 3 show that MAPPK outperforms across multiple datasets and simulated motion conditions. Our method achieves an excellent balance between artifact removal and texture structure preserva-

tion. Quantitatively, it achieves superior results in terms of peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM), demonstrating higher fidelity to the reference images. Qualitatively, MAPPK produces images with clearer tissue boundaries, more stable contrast, and finer textural representation, effectively recovering structural details that other methods tend to oversmooth or distort. These results validate the robustness and generalization capability of MAPPK in handling various types of motion artifacts, making it particularly suitable for real clinical scenarios where motion-induced distortions are unavoidable.

5.2. Qualitative evaluation of real clinical images

Next, we validate our method on clinical T2-weighted images with real artifacts, as shown in Fig. 6(A). For supervised methods, Pix2pix fails to effectively remove real motion artifacts and introduces noticeable distortions in normal tissue structures, as shown in Fig. 6(B). Among unsupervised approaches, CycleGAN not only fails to eliminate motion artifacts but even amplifies them in some regions (Fig. 6(C)). Similarly, UDDN loses its effectiveness when dealing with real motion artifacts, resulting in incomplete artifact removal and texture degradation (Fig. 6(D)). For diffusion-based methods, DR2 smooths the image when handling mild artifacts but cannot cope with severe motion artifacts (Fig. 6(E)). GDP exhibits the similar limitation that it may suppress weak artifacts but introduces artifact-like structures and fails to restore accurate textures under severe motion conditions (Fig. 6(F)). BIRD (Fig. 6(G)) completely loses its artifact removal ability when faced with real clinical artifacts, producing severely distorted textures accompanied by significant noise. PFAD (Fig. 6(H)) retains partial artifact suppression capability but distorts some texture structures, leading to unnatural appearance in certain regions. RDBM (Fig. 6(I)) improves artifact suppression compared with several previous methods, but residual artifacts and local texture inconsistency can still be observed. In

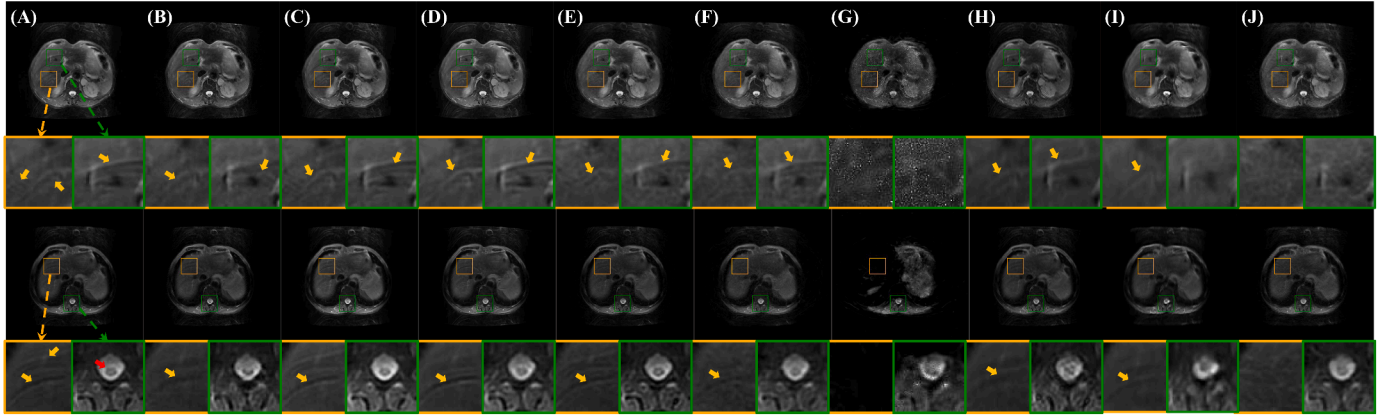


Fig. 6. Motion artifact removal results of clinical images using various methods with real artifacts: (A) Images with real motion artifacts, (B) Pix2pix, (C) CycleGAN, (D) UDDN, (E) DR2, (F) GDP, (G) BIRD (H) PFAD (I) RDBM (J) MAPPK (ours). The first and third rows show the artifact images in different regions of the abdominal cavity, respectively. And the second and fourth rows show the zoomed-in artifact regions for the first and third rows, respectively. Yellow arrow indicates the location of the artifact, while red arrow points to the critical structural regions.

contrast, the proposed MAPPK (Fig. 6(J)) demonstrates superior performance in effectively removing real motion artifacts while preserving correct anatomical textures and structural integrity. The generated images maintain realistic tissue contrast and fine-grained texture representation.

Table 4 summarizes the clinical evaluation results. All results were analyzed using the Mann-Whitney U test, with most metrics showing statistically significant differences. Our method consistently achieves superior performance, attaining the highest scores across all evaluation criteria. Furthermore, diffusion model-based approaches generally outperform GAN-based methods in clinical applications, as GANs are constrained by transformations within their finite distributions. The exception is BIRD, whose denoising network consists of only a simple linear layer, making it incapable of accurately modeling motion artifacts and resulting in distorted outputs. Overall, diffusion model-based methods demonstrate greater reliability and practical value for clinical diagnosis.

5.3. Ablation study

We further validate the effectiveness of MAPPK and its key components, including the pixel-frequency domain, alternate complementary masks, prior meanbook, residual parameter κ , dual-domain balance parameter α , and high-frequency balance parameter η . All experiments are conducted on the HCP dataset with the cut-off frequency of $\pi/20$.

5.3.1. Validation of pixel-frequency domain

We first investigate the contribution of artifact removal in the pixel domain and frequency domain separately.

Table 5 shows that artifact removal performed solely in the pixel domain only yields limited improvements in metrics like PSNR and SSIM, indicating that pixel-domain processing alone cannot fully restore fine texture details. In contrast, performing artifact removal solely in the frequency domain significantly improves PSNR and SSIM, as frequency-domain processing enhances texture information. Combining pixel- and frequency-domain processing achieves optimal overall performance, demonstrating their complementary nature: the frequency domain primarily preserves tissue texture details, while the pixel domain smoothes the output and enhances visual naturalness, thus balancing structural accuracy and image quality.

5.3.2. Validation of masks and weight

We evaluate the impact of alternate complementary masks ($\mathcal{M}$) and their weight (ω) in pixel and frequency domains.

As shown in Table 6, without masks ($\mathcal{M}$) or without weight (ω) in either domain, the restored image contains residual artifacts or incorrect

Table 5

Ablation study on the effectiveness of pixel and frequency domains.

Freq.	Pixel	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
×	×	21.77	0.522	0.082	0.410	0.749	0.135	0.919	0.171
×	✓	21.92	0.527	0.081	0.403	0.752	0.140	0.921	0.169
✓	×	25.05	0.727	0.056	0.282	0.820	0.310	0.947	0.128
✓	✓	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122

Table 6

Ablation study on the effectiveness of alternate complementary masks ($\mathcal{M}$) and the weights (ω).

$\mathcal{M}$	ω	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
Freq. Pixel									
×	×	24.55	0.707	0.060	0.298	0.814	0.286	0.943	0.134
✓	×	23.11	0.542	0.070	0.352	0.746	0.166	0.926	0.173
✓	✓	24.29	0.648	0.062	0.307	0.793	0.231	0.941	0.141
✓	×	23.26	0.560	0.069	0.346	0.754	0.174	0.929	0.169
✓	✓	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122

Table 7

Impact of prior meanbook (PM) and residual parameter (κ).

PM	κ	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
×	×	23.91	0.685	0.064	0.321	0.812	0.254	0.942	0.130
×	✓	23.80	0.713	0.065	0.325	0.812	0.273	0.938	0.133
✓	×	24.03	0.706	0.064	0.317	0.814	0.272	0.940	0.131
✓	✓	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122

high-frequency details, resulting in degradation of metrics such as PSNR and SSIM. Using $\mathcal{M}$ with proper weights ω in both domains yields the best performance, confirming that masks ($\mathcal{M}$) and their weight (ω) are essential for accurate artifact removal.

5.3.3. Validation of prior meanbook and residual parameter κ

To evaluate the prior meanbook (PM) and residual parameter κ , we conduct ablation experiments in Table 7.

Without PM or κ , the diffusion model may deviate from clean image distribution due to high-frequency artifact interference, resulting in lower metrics. Using both PM and κ achieves the highest metrics, demonstrating that PM provides stable denoising guidance, while the residual parameter κ effectively suppresses artifact interference during the reverse process.

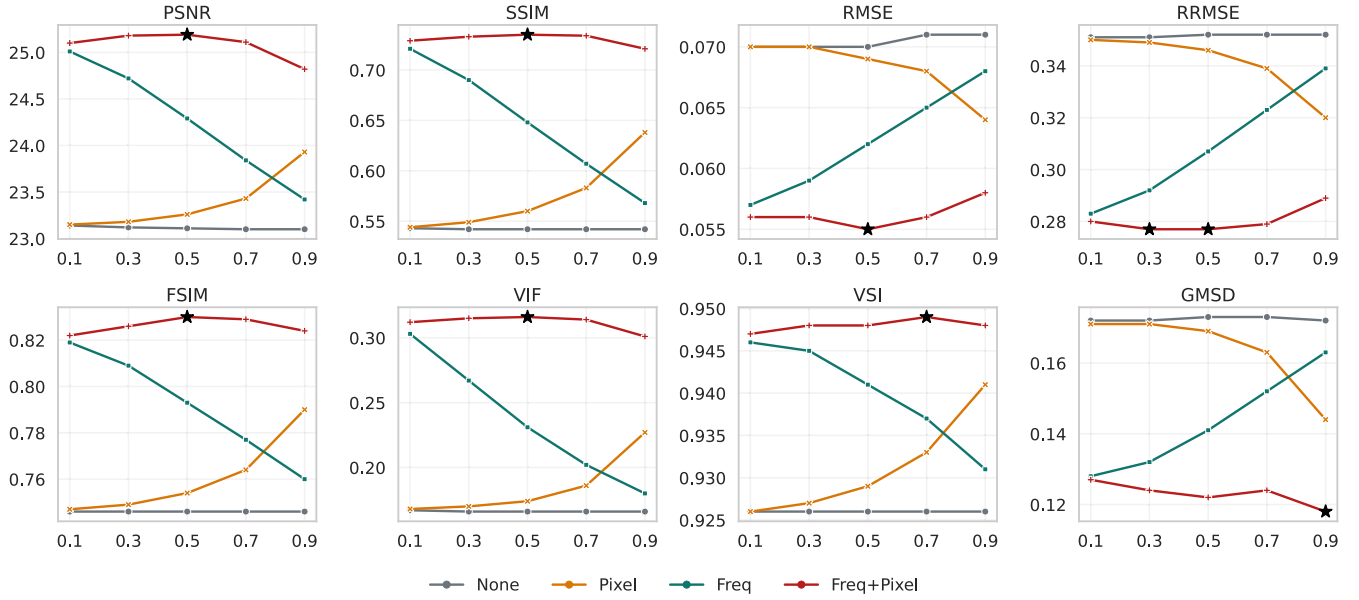


Fig. 7. The sensitivity analysis of ω and a . The horizontal axis denotes the value of a , and the vertical axis shows the corresponding metric value. The $\star$ marker highlights the best value.

Table 8
Impact of dual-domain balance parameter (a) under different values.

a	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
0.1	25.10	0.729	0.056	0.280	0.822	0.312	0.947	0.127
0.3	25.18	0.733	0.056	0.277	0.826	0.315	0.948	0.124
0.5	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122
0.7	25.11	0.734	0.056	0.279	0.829	0.314	0.949	0.124
0.9	24.82	0.721	0.058	0.289	0.824	0.301	0.948	0.118

5.3.4. Hyperparameter study on parameter a

We explore the effect of the dual-domain balance parameter a in Eq. (18) on reconstruction quality.

As presented in Table 8, when a is small (e.g., 0.1), the pixel domain plays a low role in the reconstruction process, resulting in insufficient smoothness and unnatural visual effects in the generated image, which in turn reduces overall performance. Conversely, when a is too large (e.g., 0.9), the pixel domain dominates the later stages, preventing the texture structure from being effectively preserved and also leading to a decrease in performance. The optimal range is $a = 0.5$, achieving the best performance. This shows that a effectively balances frequency and pixel domain guidance during iterative reconstruction.

5.3.5. Hyperparameter study on parameter η

Finally, we investigate the trade-off parameter η in Eq. (10) that balances the high-frequency information between diffusion prediction $\hat{x}_0$ and the prior meanbook $\hat{\mathcal{X}}_0$.

From Table 9, it can be seen that when the value of η is small (0.01–0.03), the prior meanbook does not participate enough, and thus cannot effectively guide the generation process, resulting in inadequate recovery of texture details. In contrast, when η is large (0.09), the prior meanbook contributes too much, negatively affecting the restoration of the texture and structural details of the image. The best overall performance is achieved at $\eta = 0.07$, because this moderate high-frequency weighting balances the use of model predictions with prior knowledge, effectively restoring both fine textures and precise structure.

5.3.6. Sensitivity analysis of ω and a

As shown in Fig. 7, we perform a sensitivity analysis for the mask weights ω and the dual-domain balance parameter a of γ . The results show that using either frequency-domain or pixel-domain weighting

Table 9
Impact of high-frequency balance parameter (η) under different values.

η	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
0.01	23.95	0.721	0.064	0.319	0.820	0.283	0.941	0.131
0.03	24.80	0.733	0.058	0.289	0.830	0.305	0.946	0.123
0.05	25.16	0.737	0.056	0.278	0.833	0.315	0.947	0.120
0.07	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122
0.09	25.08	0.729	0.056	0.281	0.824	0.313	0.947	0.125

Table 10
Experiment on the grid size of checkerboard mask.

Size	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD
4	25.15	0.731	0.056	0.278	0.824	0.314	0.947	0.125
16	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122
64	25.04	0.733	0.056	0.281	0.833	0.312	0.949	0.119
128	25.01	0.732	0.057	0.283	0.833	0.310	0.948	0.119

alone improves the performance compared with removing both weights, while using both achieves the best overall performance. In particular, the setting with both frequency-domain and pixel-domain weighting and $a = 0.5$ obtains the best results in metrics. These results demonstrate that the proposed dual-domain weight strategy can better balance artifact suppression and texture preservation.

5.3.7. Experiment on grid size

We use a checkerboard mask with the grid size of 16. We perform an experiment on the grid size, as shown in Table 10. Large grid size destroys the distribution of useful information and small grid size does not significantly destroy the distribution of artifact, both of them lead to poorer results, the former is worse.

5.3.8. Effect of starting timestep

To clarify the selection criterion of the starting timestep, we conduct an ablation study with different timesteps. As shown in Table 11, increasing the timestep from 100 to 200 improves the restoration performance, indicating that timestep 200 can effectively cover most motion-induced perturbation information. When the timestep is further increased, the performance becomes nearly saturated, while the inference time increases significantly. Therefore, we choose 200 as the starting timestep to balance restoration performance and inference efficiency.

Table 11
Ablation study on different starting timesteps.

Timesteps	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD	Inference Time (s)
100	25.02	0.731	0.057	0.283	0.826	0.314	0.947	0.124	5.99
200	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122	9.10
500	25.21	0.735	0.055	0.276	0.830	0.316	0.948	0.122	28.51
1000	25.22	0.735	0.055	0.276	0.830	0.316	0.948	0.122	44.08

Table 12
Hyperparameter study on the number of clusters k in the prior meanbook.

k	PSNR	SSIM	RMSE	RRMSE	FSIM	VIF	VSI	GMSD	Inference Time (s)
$\times$	23.80	0.713	0.065	0.325	0.812	0.273	0.938	0.133	7.40
10	25.10	0.731	0.056	0.279	0.829	0.314	0.948	0.123	7.91
30	25.14	0.733	0.055	0.278	0.830	0.316	0.948	0.122	8.44
50	25.19	0.735	0.055	0.277	0.830	0.316	0.948	0.122	9.10
70	25.19	0.734	0.056	0.277	0.830	0.316	0.948	0.122	9.40
90	25.13	0.733	0.056	0.279	0.829	0.314	0.948	0.123	9.67

Table 13

Average training time per epoch and inference time per slice for different methods. This result is the average result across all datasets.

Method	Training / epoch	Inference / slice
Pix2pix	2 min 11 s	0.22 s
CycleGAN	13 min 17 s	0.92 s
UDDN	38 min 42 s	1.63 s
DR2	4 min 7 s	11.64 s
GDP	4 min 7 s	10.25 s
BIRD	4 min 7 s	161.12 s
PFAD	4 min 7 s	84.83 s
RDBM	9 min 51 s	6.97 s
MAPPK (ours)	4 min 7 s	13.24 s

5.3.9. Effect of cluster number in prior meanbook

We further investigate the influence of the cluster number k in the prior meanbook. As shown in Table 12, removing the prior meanbook leads to a clear performance drop, demonstrating the effectiveness of the proposed prior guidance. Compared with the setting without the prior meanbook, introducing the prior meanbook only slightly increases the inference time. For example, when $k = 50$, the inference time increases from 7.40 s to 9.10 s, while PSNR improves from 23.80 to 25.19. When k increases from 10 to 50, the restoration performance gradually improves, suggesting that a moderate number of clusters can provide more representative prior references. However, further increasing k to 70 or 90 brings almost no additional improvement, while the inference time continues to increase due to the additional retrieval cost. Therefore, we set $k = 50$ to balance restoration performance and computational overhead. The prior meanbook retrieval introduces only limited overhead, increasing the inference time from 7.40 s without the prior meanbook to 9.10 s when $k = 50$. Overall, MAPPK provides a reasonable trade-off between restoration quality and inference efficiency.

5.4. Training and inference time comparison

We further report the average training time per epoch and inference time per slice for all methods, as shown in Table 13. In terms of training cost, Pix2pix has the shortest training time, requiring 2 min 11 s per epoch, while UDDN requires the longest training time of 38 min 42 s per epoch. For diffusion-based methods such as DR2, GDP, BIRD, PFAD, and MAPPK, the reported training time is the same because they use the same pretrained diffusion model as the generative prior. Therefore, MAPPK does not introduce additional training cost compared with these diffusion-based baselines.

For inference efficiency, GAN-based methods such as Pix2pix and CycleGAN are faster because they perform single-pass prediction. How-

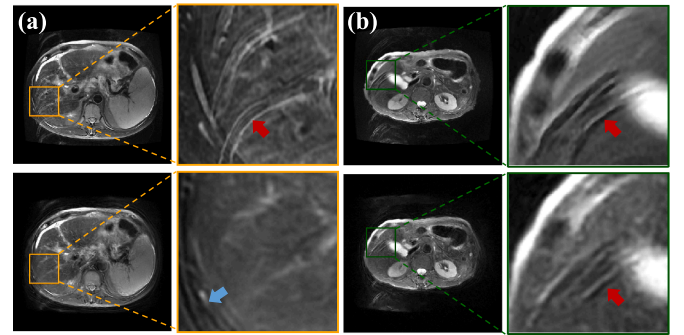


Fig. 8. Qualitative failure cases on real clinical images. (a) and (b) represent two different samples. The first row shows the original images, and the second row shows the corresponding denoised results. The red arrows indicate residual artifacts that are not completely removed, while the blue arrows indicate slight additional artifacts introduced during restoration.

ever, their restoration performance is limited when dealing with complex motion artifacts. Although MAPPK requires 13.24 s per slice due to the iterative reverse diffusion process and dual-domain guidance, it achieves better artifact suppression and anatomical structure recovery. In addition, MAPPK is much faster than BIRD and PFAD, which require 161.12 s and 84.83 s per slice, respectively. Overall, MAPPK provides a reasonable trade-off between restoration quality, training cost, and inference efficiency.

5.5. Gain compared with the original work

In this section, we compare the proposed method with our previous work to demonstrate the superiority of the current approach.

As shown in Tables 2 and 3, MAPPK consistently outperforms PFAD across all metrics on the brain, knee, and abdomen datasets. These results validate the effectiveness of newly introduced components in MAPPK, such as the Prior Meanbook and the residual parameter κ , which enhance the model's artifact removal and image reconstruction capabilities. Furthermore, visualization of the generated images (Figs. 5 and 6) confirms the effectiveness of these improvements, demonstrating clearer tissue structures and more natural anatomical details. Overall, the proposed MAPPK has been further optimized and enhanced based on our previous work.

5.6. Limitation

Although the proposed simulation strategy can generate controllable motion artifacts by introducing phase perturbations in k -space,

it may not fully cover the complexity of real clinical motion artifacts. In practice, patient motion can involve non-rigid deformation, respiratory or cardiac motion, physiological noise, and scanner-related instability. Therefore, there remains a domain gap between simulated and real-world artifacts.

The KNN-based cut-off frequency selection is constructed based on simulated artifact settings, where the degradation level can be explicitly controlled. For real clinical images, it is used as an adaptive estimation strategy to select the closest simulated degradation level. However, when real artifacts differ significantly from the simulated distributions, the selected frequency may be suboptimal, leading to incomplete artifact suppression or the introduction of slight additional artifacts. We add qualitative failure cases to better illustrate this limitation, as shown in Fig. 8.

6. Conclusion

In this paper, we propose a novel diffusion model inference framework for motion artifact removal, further extending our previous work. Specifically, by introducing alternate complementary masks in pixel-frequency domains and using the prior meanbook as the stable high-frequency guidance, we can effectively leverage partial image information to reconstruct clear images through the strong generative capability of the diffusion model. Compared with our previous work, we additionally construct a prior meanbook as the stable reference for high-frequency guidance and employ residual connections to optimize the reverse process of diffusion model, thereby enhancing artifact removal and image reconstruction quality. Experimental results demonstrate that our method consistently achieves excellent performance on both simulated and real clinical images. In summary, the proposed method builds on and extends our previous work, aiming to further mitigate the impact of motion artifacts on clinical diagnosis and assist radiologists in making more accurate assessments.

CRedit authorship contribution statement

Jiahua Xu: Writing – original draft, Methodology, Data curation; **Dawei Zhou:** Writing – original draft, Methodology, Formal analysis; **Lei Hu:** Validation, Data curation; **Jianfeng Guo:** Visualization, Data curation; **Feng Yang:** Visualization; **Zaiyi Liu:** Writing – review & editing, Supervision; **Nannan Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition; **Xinbo Gao:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the [National Natural Science Foundation of China](#) under Grants [U22A2096](#), in part by Scientific and Technological Innovation Teams in Shaanxi Province under Grant [2025RS-CXTD-011](#), in part by the Shaanxi Province Core Technology Research and Development Project under Grant [2024QY2-GJHX-11](#), in part by the Establishment Fund of the State Key Laboratory of Wakefulness/Sleep and Cognition (Chongqing Institute for Brain and Intelligence), in part by the Fundamental Research Funds for the Central Universities under Grant [QTZX26138](#), in part by Science and Technology Development Fund of Macao SAR (0059/2026/ITP1, 0054/2025/RIB2), in part by Science and Technology Development Fund of Macao SAR - International Collaborative Research (0001/2025/ALJ), in part by NSFC-FDCT Joint Scientific Research Project Fund (0004/2025/AFJ).

Data availability

The public datasets used in our paper are available in: Human Connectome Project (HCP): <https://www.humanconnectome.org/study/hcp-young-adult/data-releases> fastMRI: <https://fastmri.med.nyu.edu/> Abdominal: This study was approved by the local ethics committee, and the requirement for informed consent was waived because of the retrospective and anonymous analysis of the data (XYYYE20240082).

References

- Ahn, N., Kwon, P., Back, J., Hong, K., & Kim, S. (2023). Interactive cartoonization with controllable perceptual factors. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 16827–16835).
- Armanious, K., Tanwar, A., Abdulatif, S., Küstner, T., Gatidis, S., & Yang, B. (2020). Unsupervised adversarial correction of rigid MR motion artifacts. In *2020 IEEE 17th international symposium on biomedical imaging (ISBI)* (pp. 1494–1498). IEEE.
- Block, K. T., Uecker, M., & Frahm, J. (2007). Undersampled radial MRI with multiple coils. iterative image reconstruction using a total variation constraint. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 57(6), 1086–1098.
- Budde, J., Shajan, G., Scheffler, K., & Pohmann, R. (2014). Ultra-high resolution imaging of the human brain using acquisition-weighted imaging at 9.4 t. *NeuroImage*, 86, 592–598.
- Chen, W. K. (1990). Linear networks and systems: Algorithms and computer-aided implementations (In 2 volumes) (vol. 3). World Scientific.
- Chihouai, H., Lemkhenter, A., & Favaro, P. (2024). Blind image restoration via fast diffusion inversion. *Advances in Neural Information Processing Systems*, 37, 34513–34532.
- Chung, H., Kim, J., Yoon, J. H., Lee, J. M., & Ye, J. C. (2021). Simultaneous super-resolution and motion artifact removal in diffusion-weighted MRI using unsupervised deep learning. *arXiv:2105.00240*.
- Cruz, G., Atkinson, D., Buerger, C., Schaeffter, T., & Prieto, C. (2016). Accelerated motion corrected three-dimensional abdominal MRI using total variation regularized SENSE reconstruction. *Magnetic Resonance in Medicine*, 75(4), 1484–1498.
- Cruz, G., Atkinson, D., Henningson, M., Botnar, R. M., & Prieto, C. (2017). Highly efficient nonrigid motion-corrected 3D whole-heart coronary vessel wall imaging. *Magnetic Resonance in Medicine*, 77(5), 1894–1908.
- Dalva, Y., Altmundis, S. F., & Dundar, A. (2022). VecGAN: Image-to-image translation with interpretable latent directions. In *European conference on computer vision* (pp. 153–169). Springer.
- Dhariwal, P., & Nichol, A. (2021). Diffusion models beat GANs on image synthesis. *Advances in Neural Information Processing Systems*, 34, 8780–8794.
- Ehman, R. L., McNamara, M. T., Pallack, M., Hricak, H., & Higgins, C. B. (1984). Magnetic resonance imaging with respiratory gating: Techniques and advantages. *American Journal of Roentgenology*, 143(6), 1175–1182.
- Fei, B., Lyu, Z., Pan, L., Zhang, J., Yang, W., Luo, T., Zhang, B., & Dai, B. (2023). Generative diffusion prior for unified image restoration and enhancement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9935–9946).
- Feng, H., Wang, L., Wang, Y., Fan, H., & Huang, H. (2023). Learnability enhancement for low-light raw image denoising: A data perspective. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(1), 370–387.
- Gao, X., Xu, Z., Zhao, J., & Liu, J. (2024). Frequency-controlled diffusion model for versatile text-guided image-to-image translation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1824–1832). (vol. 38).
- Hansen, M. S., Sørensen, T. S., Arai, A. E., & Kellman, P. (2012). Retrospective reconstruction of high temporal resolution cine images from real-time MRI using iterative motion correction. *Magnetic Resonance in Medicine*, 68(3), 741–750.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.
- Ho, J., & Salimans, T. (2022). Classifier-free diffusion guidance. *arXiv:2207.12598*.
- Hu, L., Guo, X., Zhou, D., Wang, Z., Dai, L., Li, L., Li, Y., Zhang, T., Long, H., Yu, C. et al. (2024a). Development and validation of a deep learning model to reduce the interference of rectal artifacts in MRI-based prostate cancer diagnosis. *Radiology: Artificial Intelligence*, 6(2), e230362.
- Hu, L., Zhou, D., Xu, J., Lu, C., Han, C., Shi, Z., Zhu, Q., Gao, X., Wang, N., & Liu, Z. (2024b). Protecting prostate cancer classification from rectal artifacts via targeted adversarial training. *IEEE Journal of Biomedical and Health Informatics*, 28(7), 3997–4009.
- Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1125–1134).
- Jiang, C., Gao, F., Ma, B., Lin, Y., Wang, N., & Xu, G. (2023). Masked and adaptive transformer for exemplar based image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 22418–22427).
- Jin, B., & Xu, X. (2024). Machine learning coffee price predictions. *Journal of Uncertain Systems*, 17(04), 2450023.
- Jin, B., & Xu, X. (2025a). High-frequency CSI300 spot and futures price predictions via the neural network. *Journal of Uncertain Systems*, 18(04), 2550008.
- Jin, B., & Xu, X. (2025b). Late and early indica rice's price forecasts through neural networks. *International Journal of Big Data Mining for Global Warming*, 7(02), 2550005.
- Jin, B., & Xu, X. (2025c). Predictions of residential property price indices for china via machine learning models: B. Jin, X. Xu. *Quality & Quantity*, 59(Suppl 2), 1481–1513.

- Jin, B., & Xu, X. (2025d). A study of contemporaneous residential real estate price causation across major Jiangsu province cities: Methodology using vector error-correction models and directed acyclic graphs. *Economics Open*, 1, 2550008.
- Jin, B., & Xu, X. (2026). Contemporaneous causal analysis of housing prices across Guangdong's major cities: Employing vector error-correction modeling and directed acyclic graphs. *Journal of Uncertain Systems*, 2650004.
- Jin, K. H., Um, J. Y., Lee, D., Lee, J., Park, S. H., & Ye, J. C. (2017). Mri artifact correction using sparse + low-rank decomposition of annihilating filter-based hankel matrix. *Magnetic Resonance in Medicine*, 78(1), 327–340.
- Karras, T., Laine, S., & Aila, T. (2019). A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4401–4410).
- Küstner, T., Würslin, C., Schwartz, M., Martirosian, P., Gatidis, S., Brendle, C., Seith, F., Schick, F., Schwenzer, N. F., Yang, B. et al. (2017). Self-navigated 4D cartesian imaging of periodic motion in the body trunk using partial k-space compressed sensing. *Magnetic Resonance in Medicine*, 78(2), 632–644.
- Li, T., Feng, H., Wang, L., Zhu, L., Xiong, Z., & Huang, H. (2024). Stimulating diffusion model for image denoising via adaptive embedding and ensembling. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12), 8240–8257.
- Li, W., Yang, W., Hou, Y., Liu, L., Liu, Y., & Li, X. (2025). SARATR-X: Towards building a foundation model for sar target recognition. *IEEE Transactions on Image Processing*, 34, 869–884.
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 22 (140), 55.
- Liu, L., & Fieguth, P. (2012). Texture classification from random features. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(3), 574–586.
- Liu, L., Lao, S., Fieguth, P. W., Guo, Y., Wang, X., & Pietikäinen, M. (2016). Median robust extended local binary pattern for texture classification. *IEEE Transactions on Image Processing*, 25(3), 1368–1381.
- Liu, L., Sun, S., Zhi, S., Shi, F., Liu, Z., Heikkilä, J., & Liu, Y. (2025). A causal adjustment module for debiasing scene graph generation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(5), 4024–4043.
- Liu, S., Thung, K. H., Qu, L., Lin, W., Shen, D., & Yap, P. T. (2021). Learning MRI artefact removal with unpaired data. *Nature Machine Intelligence*, 3(1), 60–67.
- Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). RePaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11461–11471).
- Nichol, A., Dhariwal, P., Ramesh, A., Shyam, P., Mishkin, P., McGrew, B., Sutskever, I., & Chen, M. (2021). GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models. arXiv:2112.10741.
- Oh, G., Jung, S., Lee, J. E., & Ye, J. C. (2023). Annealed score-based diffusion model for MR motion artifact reduction. *IEEE Transactions on Computational Imaging*, 10, 43–53.
- Oh, G., Lee, J. E., & Ye, J. C. (2021). Unpaired MR motion artifact deep learning using outlier-rejecting bootstrap aggregation. *IEEE Transactions on Medical Imaging*, 40(11), 3125–3139.
- Qin, L., van Gelderen, P., Derbyshire, J. A., Jin, F., Lee, J., de Zwart, J. A., Tao, Y., & Duyn, J. H. (2009). Prospective head-movement correction for high-resolution MRI using an in-bore optical tracking system. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 62(4), 924–934.
- Qiu, D., Cheng, Y., Wong, K. K. L., Zhang, W., Yi, Z., & Wang, X. (2024a). DBSR: Quadratic conditional diffusion model for blind cardiac mri super-resolution. *IEEE Transactions on Multimedia*, 26, 11358–11371.
- Qiu, D., Shang, J., Zhang, Y., Zhang, W., Wong, K. K. L., & Yi, Z. (2026). CCDM: Continuous-time conditional diffusion model for blind cmri super-resolution. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(5), 6485–6498.
- Qiu, D., Song, Z., Zhang, W., Wong, K. K. L., & Yi, Z. (2024b). ABSR: Progressive alternate refinement for blind cardiac mri super-resolution. *IEEE Internet of Things Journal*, 11(19), 30887–30898.
- Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., & Chen, M. (2022). Hierarchical text-conditional image generation with clip latents. 1(2), 3, arXiv:2204.06125.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10684–10695).
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, october 5–9, 2015, proceedings, Part III* 18 (pp. 234–241). Springer.
- Saharia, C., Chan, W., Chang, H., Lee, C., Ho, J., Salimans, T., Fleet, D., & Norouzi, M. (2022a). Palette: Image-to-image diffusion models. In *ACM siggraph 2022 conference proceedings* (pp. 1–10).
- Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E. L., Ghasemipour, K., Gontijo Lopes, R., Karagol Ayan, B., Salimans, T. et al. (2022b). Photorealistic text-to-image diffusion models with deep language understanding. *Advances in Neural Information Processing Systems*, 35, 36479–36494.
- Schulz, J., Siegert, T., Reimer, E., Labadie, C., Maclaren, J., Herbst, M., Zaitsev, M., & Turner, R. (2012). An embedded optical tracking system for motion-corrected magnetic resonance imaging at 7T. *Magnetic Resonance Materials in Physics, Biology and Medicine*, 25, 443–453.
- Sheikh, H. R., & Bovik, A. C. (2006). Image information and visual quality. *IEEE Transactions on Image Processing*, 15(2), 430–444.
- Si, C., Huang, Z., Jiang, Y., & Liu, Z. (2023). FreeU: Free lunch in diffusion U-Net. arXiv:2309.11497.
- Song, J., Meng, C., & Ermon, S. (2020). Denoising diffusion implicit models. arXiv:2010.02502.
- Tamada, D., Kromrey, M. L., Ichikawa, S., Onishi, H., & Motosugi, U. (2020). Motion artifact reduction using a convolutional neural network for dynamic contrast enhanced MR imaging of the liver. *Magnetic Resonance in Medical Sciences*, 19(1), 64–76.
- Tumanyan, N., Geyer, M., Bagon, S., & Dekel, T. (2023). Plug-and-play diffusion features for text-driven image-to-image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1921–1930).
- Vasanawala, S. S., Alley, M. T., Hargreaves, B. A., Barth, R. A., Pauly, J. M., & Lustig, M. (2010a). Improved pediatric MR imaging with compressed sensing. *Radiology*, 256(2), 607–616.
- Vasanawala, S. S., Iwadata, Y., Church, D. G., Herfkens, R. J., & Brau, A. C. (2010b). Navigated abdominal T1-W MRI permits free-breathing image acquisition with less motion artifact. *Pediatric Radiology*, 40, 340–344.
- Wang, H., Zhang, J., Chen, H., Guo, H., Wang, D., Ma, J., & Du, B. (2026). Residual diffusion bridge model for image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8375–8386).
- Wang, T. C., Liu, M. Y., Zhu, J. Y., Tao, A., Kautz, J., & Catanzaro, B. (2018). High-resolution image synthesis and semantic manipulation with conditional gans. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 8798–8807).
- Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4), 600–612.
- Wang, Z., Zhang, Z., Zhang, X., Zheng, H., Zhou, M., Zhang, Y., & Wang, Y. (2023). DR2: Diffusion-based robust degradation remover for blind face restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1704–1713).
- White, M. J., Hawkes, D. J., Melbourne, A., Collins, D. J., Coolens, C., Hawkins, M., Leach, M. O., & Atkinson, D. (2009). Motion artifact correction in free-breathing abdominal MRI using overlapping partial samples to recover image deformations. *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine*, 62(2), 440–449.
- Wu, B., Li, C., Zhang, J., Lai, H., Feng, Q., & Huang, M. (2023). Unsupervised dual-domain disentangled network for removal of rigid motion artifacts in MRI. *Computers in Biology and Medicine*, 165, 107373.
- Xu, J., Zhou, D., Hu, L., Guo, J., Yang, F., Liu, Z., Wang, N., & Gao, X. (2025). Motion artifact removal in pixel-frequency domain via alternate masks and diffusion model. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 8878–8886). (vol. 39).
- Xu, X. (2020). Corn cash price forecasting. *American Journal of Agricultural Economics*, 102(4), 1297–1320.
- Xu, X., & Zhang, Y. (2021a). Individual time series and composite forecasting of the chinese stock index. *Machine Learning with Applications*, 5, 100035.
- Xu, X., & Zhang, Y. (2021b). Network analysis of corn cash price comovements. *Machine Learning with Applications*, 6, 100140.
- Xu, X., & Zhang, Y. (2023). Network analysis of housing price comovements of a hundred chinese cities. *National Institute Economic Review*, 264, 110–128.
- Xue, W., Zhang, L., Mou, X., & Bovik, A. C. (2013). Gradient magnitude similarity deviation: A highly efficient perceptual image quality index. *IEEE Transactions on Image Processing*, 22(2), 684–695.
- Yang, Z., Zhang, C., & Xie, L. (2013). Sparse MRI for motion correction. In *2013 IEEE 10th international symposium on biomedical imaging* (pp. 962–965). IEEE.
- Zaitsev, M., Dold, C., Sakas, G., Hennig, J., & Speck, O. (2006). Magnetic resonance imaging of freely moving objects: Prospective real-time motion correction using an external optical motion tracking system. *NeuroImage*, 31(3), 1038–1050.
- Zaitsev, M., Maclaren, J., & Herbst, M. (2015). Motion artifacts in MRI: A complex problem with many partial solutions. *Journal of Magnetic Resonance Imaging*, 42(4), 887–901.
- Zbontar, J., Knoll, F., Sriram, A., Murrell, T., Huang, Z., Muckley, M. J., Defazio, A., Stern, R., Johnson, P., Bruno, M. et al. (2018). fastMRI: An open dataset and benchmarks for accelerated MRI. arXiv:1811.08839.
- Zhang, J., Pan, X., & Lv, T. (2021). Unsupervised MRI images denoising via decoupled expression. In *International conference on wireless communications, networking and applications* (pp. 769–777). Springer.
- Zhang, L., Rao, A., & Agrawala, M. (2023). Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3836–3847).
- Zhang, L., Shen, Y., & Li, H. (2014). VSI: A visual saliency-induced index for perceptual image quality assessment. *IEEE Transactions on Image Processing*, 23(10), 4270–4281.
- Zhang, L., Zhang, L., Mou, X., & Zhang, D. (2011). FSIM: A feature similarity index for image quality assessment. *IEEE Transactions on Image Processing*, 20(8), 2378–2386.
- Zhang, Y., Li, W., Jia, W., Zhang, M., Tao, R., & Liang, S. (2025a). Cross-domain hyperspectral image classification based on bi-directional domain adaptation. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(12), 12038–12051.
- Zhang, Y., Li, W., Zhang, M., Han, J., Tao, R., & Liang, S. (2025b). SpectralX: Parameter-efficient domain generalization for spectral remote sensing foundation models. arXiv:2508.01731.
- Zhang, Y., Liang, S., Li, W., Ma, H., Xu, J., Ma, Y., Xie, J., Li, W., Zhang, M., Tao, R. et al. (2025c). UniTS: Unified time series generative model for remote sensing. arXiv e-prints, (pp. arXiv–2512).
- Zhou, D., Wang, N., Han, B., Liu, T., & Gao, X. (2025a). Defending against adversarial examples via modeling adversarial noise. *International Journal of Computer Vision*, 133(9), 5920–5937.
- Zhou, D., Wang, N., Liu, T., & Gao, X. (2025b). Improving adversarial training from the perspective of class-flipping distribution. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(6), 4330–4342.
- Zhu, J. Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision* (pp. 2223–2232).